

Chapter 7

Determinants

Solutions

SECTION - A

Objective Type Questions (One option is correct)

[Introduction and Properties of Determinants]

1. Let $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, then $|2A|$ is equal to

(1) $4\cos 2\theta$ (2) 1 (3) 2 (4) 4

Sol. Answer (4)

$$|A| = \cos^2\theta + \sin^2\theta = 1, |2A| = 2^2 \cdot |A| = 4$$

2. If ω is non-real complex cube root of unity, then $\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & \omega & 1 \end{vmatrix}$ is equal to

(1) 0 (2) 1 (3) 3 (4) 2

Sol. Answer (1)

$$A = \begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & \omega & 1 \end{vmatrix} = \begin{vmatrix} 1+\omega+\omega^2 & \omega & \omega^2 \\ \omega+\omega^2+1 & \omega^2 & 1 \\ \omega^2+\omega+1 & \omega & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & \omega & \omega^2 \\ 0 & \omega^2 & 1 \\ 0 & \omega & 1 \end{vmatrix} = 0$$

3. The value of the determinant $\begin{vmatrix} 1 & x & x^3 \\ 1 & y & y^3 \\ 1 & z & z^3 \end{vmatrix}$ is equal to

(1) $(x-y)(y-z)(z-x)$ (2) $(x-y)(y-z)(z-x)(x+y+z)$
(3) $(x+y+z)$ (4) $(x-y)(y-z)(z-x)(xy+yz+zx)$

Sol. Answer (2)

$$\Delta = \begin{vmatrix} 1 & x & x^3 \\ 1 & y & y^3 \\ 1 & z & z^3 \end{vmatrix}$$

$$R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_3$$

$$\Delta = \begin{vmatrix} 0 & x-y & (x-y)(x^2+xy+y^2) \\ 0 & y-z & (y-z)(y^2+yz+z^2) \\ 1 & z & z^3 \end{vmatrix}$$

$$= (x-y)(y-z)(z-x)(x+y+z)$$

4. If A is 3×3 matrix and $|A| = 4$, then $|A^{-1}|$ is equal to

(1) $\frac{1}{4}$

(2) $\frac{1}{16}$

(3) 4

(4) 2

Sol. Answer (1)

$$|A^{-1}| = \frac{1}{|A|} = \frac{1}{4}$$

5. If A , B and C are three square matrices of the same order such that $A = B + C$, then $\det A$ is equal to

(1) $\det A + \det B$

(2) $\det B$

(3) $\det(A)$

(4) $\det(B + C)$

Sol. Answer (4)

$$A = B + C$$

For addition, there is not specific rule.

6. If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, then value of $\begin{vmatrix} 2a_1+3b_1+4c_1 & b_1 & c_1 \\ 2a_2+3b_2+4c_2 & b_2 & c_2 \\ 2a_3+3b_3+4c_3 & b_3 & c_3 \end{vmatrix}$ is equal to

(1) Δ^2

(2) 4Δ

(3) Δ

(4) 2Δ

Sol. Answer (4)

7. Let $ax^3 + bx^2 + cx + d = \begin{vmatrix} 3x & x+1 & x-1 \\ x-3 & -2x & x+2 \\ x+3 & x-4 & 5x \end{vmatrix}$, then the value of d is

(1) 5

(2) 0

(3) -6

(4) 4

Sol. Answer (3)Put $x = 0$

$$d = \begin{vmatrix} 0 & 1 & -1 \\ -3 & 0 & 2 \\ 3 & -4 & 0 \end{vmatrix}$$

$$\Rightarrow d = -6$$

8. If $A + B + C = \pi$, then value of

$$\begin{vmatrix} \sin(A+B+C) & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ \cos(A+B) & -\tan A & 0 \end{vmatrix} \text{ is}$$

(1) 0

(2) 1

(3) $2\sin B \cdot \tan A \cos C$ (4) $2\sin A \sin B \cdot \sin C$

Sol. Answer (1)

$$\Delta = \begin{vmatrix} \sin \pi & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ \cos(\pi - C) & -\tan A & 0 \end{vmatrix}$$

$$\Delta = \begin{vmatrix} 0 & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ -\cos C & -\tan A & 0 \end{vmatrix}$$

= 0 (Skew symmetric matrix of odd order)

9. The value of θ lying between 0 and $\frac{\pi}{2}$ and satisfying the equation $\begin{vmatrix} 1 + \sin^2 \theta & \cos^2 \theta & 4 \sin 4\theta \\ \sin^2 \theta & 1 + \cos^2 \theta & 4 \sin 4\theta \\ \sin^2 \theta & \cos^2 \theta & 1 + 4 \sin 4\theta \end{vmatrix} = 0$ are

(1) $\frac{5\pi}{24}, \frac{3\pi}{24}$ (2) $\frac{7\pi}{24}, \frac{5\pi}{24}$ (3) $\frac{7\pi}{24}, \frac{11\pi}{24}$ (4) $\frac{\pi}{24}, \frac{11\pi}{24}$

Sol. Answer (3)

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$$

$$\Delta = \begin{vmatrix} 1 + \sin^2 \theta & \cos^2 \theta & 4 \sin 4\theta \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = 0$$

$$C_1 \rightarrow C_1 + C_2 + C_3$$

$$\Rightarrow \begin{vmatrix} 2 + 4 \sin 4\theta & \cos^2 \theta & 4 \sin 4\theta \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 0$$

$$\Rightarrow 4 \sin 4\theta = -2$$

$$\Rightarrow \sin 4\theta = -\frac{1}{2} = \sin\left(-\frac{\pi}{6}\right)$$

$$\Rightarrow \theta = \frac{7\pi}{24}, \frac{11\pi}{24}$$

10. The number of real roots of the equation $\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$ is

(1) 0

(2) 1

(3) 2

(4) 3

Sol. Answer (4)

$$\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$$

$$\Rightarrow -5(x-2)(x+3)(x-1) = 0$$

$$\Rightarrow x = 2, -3, 1$$

11. If $\begin{vmatrix} 4 & -4 & 0 \\ a & b+4 & c \\ a & b & c+4 \end{vmatrix} = 0$, then $a + b + c$ is equal to

(1) 41

(2) 116

(3) 628

(4) -4

Sol. Answer (4)

$$C_1 \rightarrow C_1 + C_2 + C_3$$

$$\begin{vmatrix} 0 & -4 & 0 \\ a+b+c+4 & b+4 & c \\ a+b+c+4 & b & c+4 \end{vmatrix} = 0$$

$$\Rightarrow (a + b + c + 4) = 0$$

$$\Rightarrow a + b + c = -4$$

12. The equation $\begin{vmatrix} x-2 & 3 & 1 \\ 4x-2 & 10 & 4 \\ 2x-1 & 5 & 1 \end{vmatrix} = 0$ is satisfied by

(1) $x = -2$ (2) $x = -5$ (3) $x = -7$ (4) $x = -9$ **Sol.** Answer (3)

$$\begin{vmatrix} x-2 & 3 & 1 \\ 4x-2 & 10 & 4 \\ 2x-1 & 5 & 1 \end{vmatrix} = 0$$

$$(x-2)(10-20) - (4x-2)(3-5) + (2x-1)(12-10) = 0$$

$$= -10(x-2) + 2(4x-2) + 2(2x-1) = 0$$

$$= -10x + 20 + 8x - 4 + 4x - 2 = 0$$

$$\Rightarrow 2x + 14 = 0$$

$$\Rightarrow x = -7$$

13. $\begin{vmatrix} x & 4 & y+z \\ y & 4 & z+x \\ z & 4 & x+y \end{vmatrix}$ is equal to

- (1) 4 (2) $x + y + z$ (3) xyz (4) 0

Sol. Answer (4)

$$R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_1$$

$$\begin{vmatrix} x-y & 0 & y-x \\ y-z & 0 & z-y \\ z & 4 & x+y \end{vmatrix}$$

$$= -4((x-y)(z-y) - (y-z)(y-x)) = 0$$

14. The value of the determinant $\begin{vmatrix} 1 & a & a^2 - bc \\ 1 & b & b^2 - ca \\ 1 & c & c^2 - ab \end{vmatrix}$ is

- (1) 0 (2) 1 (3) 2 (4) 3

Sol. Answer (1)

Given $\begin{vmatrix} 1 & a & a^2 - bc \\ 1 & b & b^2 - ca \\ 1 & c & c^2 - ab \end{vmatrix}$

$$R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_3$$

$$= (a-b)(b-c) \begin{vmatrix} 0 & 1 & a+b+c \\ 0 & 1 & a+b+c \\ 1 & c & c^2 - ab \end{vmatrix}$$

$$= 0 \quad [\because R_1 \text{ and } R_2 \text{ are identical}]$$

15. The roots of the equation $\begin{vmatrix} x-1 & 1 & 1 \\ 1 & x-1 & 1 \\ 1 & 1 & x-1 \end{vmatrix} = 0$ are

- (1) 1, 2 (2) -1, 2 (3) 1, -2 (4) -1, -2

Sol. Answer (2)

$$C_1 \rightarrow C_1 + C_2 + C_3$$

$$\Delta = \begin{vmatrix} x-1 & 1 & 1 \\ 1 & x-1 & 1 \\ 1 & 1 & x-1 \end{vmatrix}$$

$$= \begin{vmatrix} x+1 & 1 & 1 \\ x+1 & x-1 & 1 \\ x+1 & 1 & x-1 \end{vmatrix}$$

$$= (x+1) \begin{vmatrix} 1 & 1 & 1 \\ 1 & x-1 & 1 \\ 1 & 1 & x-1 \end{vmatrix}$$

$$= (x+1) \begin{vmatrix} 0 & -x+2 & 0 \\ 0 & x-2 & 2-x \\ 1 & 1 & x-1 \end{vmatrix}$$

$$= -(x+1)(x-2)^2$$

Roots are $-1, 2, 2$

16. The value of the determinant $\begin{vmatrix} 1 & \log_b a \\ \log_a b & 1 \end{vmatrix}$ is equal to

- (1) 1 (2) $\log_a b$ (3) $\log_b a$ (4) 0

Sol. Answer (4)

$$\begin{aligned} \Delta &= 1 \cdot 1 - \log_b a \times \log_a b \\ &= 1 - 1 = 0 \end{aligned}$$

17. $\begin{vmatrix} 11+a & c & 1+bc \\ 11+a & d & 1+bd \\ 11+a & e & 1+be \end{vmatrix}$ is equal to

- (1) 1 (2) 0 (3) 3 (4) $a + b + c$

Sol. Answer (2)

$$\begin{vmatrix} 0 & c-d & b(c-d) \\ 0 & d-e & b(d-e) \\ 11+a & e & 1+be \end{vmatrix}$$

$$= (c-d)(d-e) \begin{vmatrix} 0 & 1 & b \\ 0 & 1 & b \\ 11+a & e & 1+be \end{vmatrix} = 0 \quad (\text{as } R_1 = R_2)$$

18. If $\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = ka^2b^2c^2$, then k is equal to

- (1) 2 (2) 4 (3) -4 (4) 8

Sol. Answer (2)

Put $a = b = c = 1$

$$\begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = k$$

$$k = -1(0) - 1(-2) + 1(2) = 4$$

19. Let $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix}$, then $f(100)$ is equal to

(1) 0

(2) 1

(3) 100

(4) -100

Sol. Answer (1)

$$f(x) = x(x+1) \begin{vmatrix} 1 & 1 & 1 \\ 2x & x-1 & x \\ 3x(x-1) & (x-1)(x-2) & x(x-1) \end{vmatrix}$$

$$C_1 \rightarrow C_1 - C_2, C_2 \rightarrow C_2 - C_3$$

$$x(x+1) \begin{vmatrix} 0 & 0 & 1 \\ x+1 & -1 & x \\ 2(x-1)(x+1) & -2(x-1) & x(x-1) \end{vmatrix}$$

$$= x(x+1)(-2(x^2-1) + 2(x^2-1)) = 0$$

$$\therefore f(100) = 0$$

20. If $s = (a + b + c)$, then value of $\begin{vmatrix} s+c & a & b \\ c & s+a & b \\ c & a & s+b \end{vmatrix}$ is

(1) $2s^2$ (2) $2s^3$ (3) s^3 (4) $3s^3$ **Sol.** Answer (2)

$$\Delta = \begin{vmatrix} s+c & a & b \\ c & s+a & b \\ c & a & s+b \end{vmatrix}$$

$$C_1 \rightarrow C_1 + C_2 + C_3$$

$$\Delta = \begin{vmatrix} s+a+b+c & a & b \\ s+a+b+c & s+a & b \\ s+a+b+c & a & s+b \end{vmatrix}$$

$$= (s+a+b+c) \begin{vmatrix} 1 & a & b \\ 1 & s+a & b \\ 1 & a & s+b \end{vmatrix}$$

$$= (2s) \begin{vmatrix} 0 & -s & 0 \\ 0 & s & -s \\ 1 & a & s+b \end{vmatrix}$$

$$= (2s)(s^2 - c)$$

$$= 2s^3$$

21. The number of distinct real roots of $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ in the interval $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$ is

(1) 0

(2) 2

(3) 1

(4) 3

Sol. Answer (3)

$$\Delta = \begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix}$$

$$C_1 \rightarrow C_1 + C_2 + C_3$$

$$= \begin{vmatrix} \sin x + 2\cos x & \cos x & \cos x \\ \sin x + 2\cos x & \sin x & \cos x \\ \sin x + 2\cos x & \cos x & \sin x \end{vmatrix}$$

$$= (\sin x + 2\cos x) \begin{vmatrix} 1 & \cos x & \cos x \\ 1 & \sin x & \cos x \\ 1 & \cos x & \sin x \end{vmatrix}$$

$$= (\sin x + 2\cos x) \begin{vmatrix} 0 & \cos x - \sin x & 0 \\ 0 & \sin x - \cos x & \cos x - \sin x \\ 1 & \cos x & \sin x \end{vmatrix}$$

$$= (\sin x + 2\cos x)(\cos x - \sin x)^2 \begin{vmatrix} 0 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & \cos x & \sin x \end{vmatrix}$$

$$= (\sin x + 2\cos x)(\cos x - \sin x)^2 = 0$$

$$\Rightarrow \tan x = -2 \text{ or } \sin x = \cos x$$

$$x = \frac{\pi}{4} \text{ is only solution}$$

$$\therefore \text{Number of solution} = \text{one}$$

22. $\begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (b^x + b^{-x})^2 & (b^x - b^{-x})^2 & 1 \\ (c^x + c^{-x})^2 & (c^x - c^{-x})^2 & 1 \end{vmatrix}$ is equal to

(1) 0

(2) $2abc$ (3) $a^2b^2c^2$ (4) abc

Sol. Answer (1)

$$C_1 \rightarrow C_1 - C_2$$

$$\begin{vmatrix} 4 & (a^x - a^{-x})^2 & 1 \\ 4 & (b^x - b^{-x})^2 & 1 \\ 4 & (c^x - c^{-x})^2 & 1 \end{vmatrix}$$

$$= 0 \text{ (as } C_1 \text{ and } C_3 \text{ are proportional)}$$

23. The determinant

$$D = \begin{vmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) & \cos 2\beta \\ \sin \alpha & \cos \alpha & \sin \beta \\ -\cos \alpha & \sin \alpha & \cos \beta \end{vmatrix} \text{ is independent of}$$

- (1) α (2) β (3) α and β (4) Neither α nor β

Sol. Answer (1)

$$\begin{aligned} D &= \cos(\alpha + \beta)(\cos \alpha \cos \beta - \sin \alpha \sin \beta) + \sin(\alpha + \beta)(\sin \alpha \cos \beta + \cos \alpha \sin \beta) + \cos 2\beta(\sin^2 \alpha + \cos^2 \alpha) \\ &= \cos(\alpha + \beta) \cos(\alpha + \beta) + \sin(\alpha + \beta) \sin(\alpha + \beta) + \cos 2\beta \cdot 1 \\ &= \cos 2\beta \end{aligned}$$

24. The roots of the equation $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0$ are

- (1) $-1, -2$ (2) $-1, 2$ (3) $1, -2$ (4) $1, 2$

Sol. Answer (2)

$$\begin{aligned} \Delta &= \begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} \\ &= \begin{vmatrix} 0 & 6 & 15 \\ 0 & -2-2x & 5(1-x^2) \\ 1 & 2x & 5x^2 \end{vmatrix} \\ &= 30(1-x^2) + 30(1+x) = 0 \\ \Rightarrow 1-x^2 + 1+x &= 0 \\ \Rightarrow x^2 - x - 2 &= 0 \\ \Rightarrow x &= 2, -1 \end{aligned}$$

25. If a, b, c are sides of the $\triangle ABC$, $\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$, then value of $\sin^2 A + \sin^2 B + \sin^2 C$ is equal to

- (1) $\frac{4}{9}$ (2) $\frac{3}{2}$ (3) $\frac{9}{4}$ (4) $\frac{3\sqrt{3}}{2}$

Sol. Answer (3)

$$\begin{aligned} \begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} &= 0 \\ \Rightarrow \begin{vmatrix} 0 & a-c & b-a \\ 0 & c-b & a-c \\ 1 & b & c \end{vmatrix} &= 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow (a-c)^2 - (b-a)(c-b) &= 0 \\ \Rightarrow a^2 + c^2 - 2ac - (bc - b^2 - ac + ab) & \\ \Rightarrow a^2 + b^2 + c^2 - ab - bc - ca &= 0 \\ \Rightarrow a = b = c \end{aligned}$$

$$\therefore \sin^2 A + \sin^2 B + \sin^2 C = \frac{3}{4} + \frac{3}{4} + \frac{3}{4} = \frac{9}{4}$$

26. The parameter, on which the value of the determinant $\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$ does not depend

upon, is

- (1) a (2) p (3) d (4) x

Sol. Answer (2)

$$\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$$

$$\begin{aligned} &= (\cos px \cdot \sin(p+d)x - \sin px \cdot \cos(p+d)x) - a(\cos(p-d)x \cdot \sin(p+d)x - \sin(p-d)x \cdot \cos(p+d)x) + a^2(\cos(p-d)x \cdot \sin px - \sin(p-d)x \cdot \cos px) \\ &= \sin dx - a \sin 2dx + a^2 \sin dx \end{aligned}$$

27. If $A^2 = 8A + kI$ where $A = \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix}$, then k is

- (1) 7 (2) -7 (3) 1 (4) -1

Sol. Answer (2)

$$A^2 = \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -8 & 49 \end{bmatrix}$$

$$8A + kI = \begin{bmatrix} 8 & 0 \\ -8 & 56 \end{bmatrix} + \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} = \begin{bmatrix} 8+k & 0 \\ -8 & 56+k \end{bmatrix}$$

Clearly, $k = -7$

28. The value of the determinant $\begin{vmatrix} {}^5C_0 & {}^5C_3 & 14 \\ {}^5C_1 & {}^5C_4 & 1 \\ {}^5C_2 & {}^5C_5 & 1 \end{vmatrix}$ is

- (1) 0 (2) $-(6!)$ (3) 80 (4) -576

Sol. Answer (4)

$$\begin{vmatrix} {}^5C_0 & {}^5C_3 & 14 \\ {}^5C_1 & {}^5C_4 & 1 \\ {}^5C_2 & {}^5C_5 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 10 & 14 \\ 5 & 5 & 1 \\ 10 & 1 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 16 & 16 & 16 \\ 5 & 5 & 1 \\ 10 & 1 & 1 \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$= 16 \begin{vmatrix} 1 & 1 & 1 \\ 5 & 5 & 1 \\ 10 & 1 & 1 \end{vmatrix}$$

$$= 16 \begin{vmatrix} 0 & 0 & 1 \\ 0 & 4 & 1 \\ 9 & 0 & 1 \end{vmatrix}$$

$$C_1 \rightarrow C_1 - C_2,$$

$$C_2 \rightarrow C_2 - C_3$$

$$= 16(0 - 36)$$

$$= -16 \times 36$$

29. If $\Delta_1 = \begin{vmatrix} 10 & 4 & 3 \\ 17 & 7 & 4 \\ 4 & -5 & 7 \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} 4 & x+5 & 3 \\ 7 & x+12 & 4 \\ -5 & x-1 & 7 \end{vmatrix}$ such that $\Delta_1 + \Delta_2 = 0$, then

(1) $x = 5$

(2) x has no real value

(3) $x = 0$

(4) $x = 1$

Sol. Answer (1)

$$\Delta_1 = \begin{vmatrix} 10 & 4 & 3 \\ 17 & 7 & 4 \\ 4 & -5 & 7 \end{vmatrix}$$

$$= - \begin{vmatrix} 4 & 10 & 3 \\ 7 & 17 & 4 \\ -5 & 4 & 7 \end{vmatrix}$$

...(i)

If $\Delta_1 + \Delta_2 = 0$, then $\Delta_1 = -\Delta_2$

...(ii)

By equation (i) and (ii),

$$\begin{vmatrix} 4 & x+5 & 3 \\ 7 & x+12 & 4 \\ -5 & x-1 & 7 \end{vmatrix} = \begin{vmatrix} 4 & 10 & 3 \\ 7 & 17 & 4 \\ -5 & 4 & 7 \end{vmatrix}$$

$$\Rightarrow x = 5$$

30. The value of $\begin{vmatrix} x & x^2 - yz & 1 \\ y & y^2 - zx & 1 \\ z & z^2 - xy & 1 \end{vmatrix}$ is

(1) 1

(2) -1

(3) 0

(4) $-xyz$

Sol. Answer (3)

$$\Delta = \begin{vmatrix} x & x^2 - yz & 1 \\ y & y^2 - zx & 1 \\ z & z^2 - xy & 1 \end{vmatrix} \text{ can be resolved as sum by two determinants as,}$$

$$\Delta = \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & -yz & 1 \\ y & -zx & 1 \\ z & -xy & 1 \end{vmatrix}$$

$$= \Delta_1 + \Delta_2 \text{ (say)}$$

$$\text{where, } \Delta_2 = \begin{vmatrix} x & -yz & 1 \\ y & -zx & 1 \\ z & -xy & 1 \end{vmatrix}$$

$$= \frac{1}{xyz} \begin{vmatrix} x^2 & -xyz & x \\ y^2 & -xyz & y \\ z^2 & -xyz & z \end{vmatrix} \text{ (multiplying } R_1 \text{ by } x, R_2 \text{ by } y \text{ and } R_3 \text{ by } z)$$

$$= \frac{-xyz}{xyz} \begin{vmatrix} x^2 & 1 & x \\ y^2 & 1 & y \\ z^2 & 1 & z \end{vmatrix} \text{ (taking common } -xyz \text{ from } C_2)$$

$$= -(-1) \begin{vmatrix} x^2 & x & 1 \\ y^2 & y & 1 \\ z^2 & z & 1 \end{vmatrix}$$

$$= -(-1)(-1) \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix}$$

$$= -\Delta_1$$

$$\Rightarrow \Delta = \Delta_1 + \Delta_2 = 0$$

31. The value of $\begin{vmatrix} i^m & i^{m+1} & i^{m+2} \\ i^{m+5} & i^{m+4} & i^{m+3} \\ i^{m+6} & i^{m+7} & i^{m+8} \end{vmatrix}$, when $i = \sqrt{-1}$, is

(1) 1 if m is a multiple of 4

(2) 0 for all real m

(3) $-i$ if m is a multiple of 3

(4) 2

Sol. Answer (2)

$$\Delta = \begin{vmatrix} i^m & i^{m+1} & i^{m+2} \\ i^{m+5} & i^{m+4} & i^{m+3} \\ i^{m+6} & i^{m+7} & i^{m+8} \end{vmatrix}$$

$$= i^m \cdot i^{m+3} \cdot i^{m+6} \begin{vmatrix} 1 & i & i^2 \\ i^2 & i & 1 \\ 1 & i & i^2 \end{vmatrix}$$

$$= 0 \quad (\because R_1 \text{ \& } R_3 \text{ are identical})$$

32. The value of the determinant $\begin{vmatrix} -(2^5 + 1)^2 & 2^{10} - 1 & \frac{1}{2^5 - 1} \\ 2^{10} - 1 & -(2^5 - 1)^2 & \frac{1}{2^5 + 1} \\ \frac{1}{2^5 - 1} & \frac{1}{2^5 + 1} & -\frac{1}{(2^{10} - 1)^2} \end{vmatrix}$ is

(1) 0

(2) 1

(3) 2

(4) 4

Sol. Answer (4)

Taking $2^5 + 1 = a$ and $2^5 - 1 = b$, then $2^{10} - 1 = (2^5 + 1)(2^5 - 1) = ab$, therefore the given determinant equals.

$$\Delta = \begin{vmatrix} -a^2 & ab & \frac{1}{b} \\ ab & -b^2 & \frac{1}{a} \\ \frac{1}{b} & \frac{1}{a} & -\frac{1}{a^2 b^2} \end{vmatrix}$$

Multiplying R_1 with b , R_2 with a and R_3 with $a^2 b^2$

$$\Delta = \frac{1}{a^3 b^3} \begin{vmatrix} -ba^2 & ab^2 & 1 \\ ba^2 & -ab^2 & 1 \\ ba^2 & ab^2 & -1 \end{vmatrix}$$

$$= \frac{1}{a^3 b^3} \begin{vmatrix} 0 & 0 & 2 \\ ba^2 & -ab^2 & 1 \\ ba^2 & ab^2 & -1 \end{vmatrix}$$

$$= \frac{2}{a^3 b^3} \begin{vmatrix} ba^2 & -ab^2 \\ ba^2 & ab^2 \end{vmatrix}$$

$$= \frac{2}{a^3 b^3} (a^3 b^3) \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix}$$

$$= 4$$

33. If the determinant $\begin{vmatrix} \cos 2x & \sin^2 x & \cos 4x \\ \sin^2 x & \cos 2x & \cos^2 x \\ \cos 4x & \cos^2 x & \cos 2x \end{vmatrix}$ is expanded in powers of $\sin x$ then the constant term in the

expansion is

(1) 1

(2) 2

(3) -1

(4) 0

Sol. Answer (3)

For constant term, we can substitute $x = 0$.

$$\therefore \text{The constant term} = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= -1$$

34. If α, β are non-real numbers satisfying $x^3 - 1 = 0$ then the value of $\begin{vmatrix} \lambda+1 & \alpha & \beta \\ \alpha & \lambda+\beta & 1 \\ \beta & 1 & \lambda+\alpha \end{vmatrix}$ is equal to
- (1) 0 (2) λ^3 (3) $\lambda^3 + 1$ (4) 4

Sol. Answer (2)

$$x^3 - 1 = 0 \Rightarrow x = 1, \omega, \omega^2$$

$$\therefore \alpha = \omega, \beta = \omega^2 \Rightarrow 1 + \alpha + \beta = 0 \text{ and } \alpha\beta = 1$$

$$\Delta = \begin{vmatrix} \lambda+1 & \alpha & \beta \\ \alpha & \lambda+\beta & 1 \\ \beta & 1 & \lambda+\alpha \end{vmatrix}$$

$$= \begin{vmatrix} \lambda+1+\alpha+\beta & \alpha & \beta \\ \lambda+1+\alpha+\beta & \lambda+\beta & 1 \\ \lambda+1+\alpha+\beta & 1 & \lambda+\alpha \end{vmatrix} \quad C_1 \rightarrow C_1 + C_2 + C_3$$

$$= \lambda \begin{vmatrix} 1 & \alpha & \beta \\ 1 & \lambda+\beta & 1 \\ 1 & 1 & \lambda+\alpha \end{vmatrix} \quad (\because 1+\alpha+\beta=0)$$

$$= \lambda \begin{vmatrix} 1 & \alpha & \beta \\ 0 & \lambda+\beta-\alpha & 1-\beta \\ 0 & 1-\alpha & \lambda+\alpha-\beta \end{vmatrix} \quad \begin{matrix} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{matrix}$$

$$\begin{aligned} &= \lambda[(\lambda + \alpha - \beta)(\lambda + \beta - \alpha) - (1 - \alpha)(1 - \beta)] \\ &= \lambda[\lambda^2 - (\alpha - \beta)^2 - (1 - \alpha - \beta + \alpha\beta)] \\ &= \lambda[\lambda^2 - (\alpha + \beta)^2 + 4\alpha\beta - (1 - (\alpha + \beta) + \alpha\beta)] \\ &= \lambda[\lambda^2 - (-1)^2 + 4 - (1 - (-1) + 1)] \\ &= \lambda(\lambda^2 - 1 + 4 - 3) \\ &= \lambda^3 \end{aligned}$$

35. The value of $\sum_{n=1}^N U_n$ if $U_n = \begin{vmatrix} n & 1 & 5 \\ n^2 & 2N+1 & 2N+1 \\ n^3 & 3N^2 & 3N \end{vmatrix}$ is

- (1) 0 (2) 1 (3) -1 (4) 2

Sol. Answer (1)

$$\sum_{n=1}^N U_n = \begin{vmatrix} \sum_{n=1}^N n & 1 & 5 \\ \sum_{n=1}^N n^2 & 2N+1 & 2N+1 \\ \sum_{n=1}^N n^3 & 3N^2 & 3N \end{vmatrix}$$

$$\begin{aligned}
 &= \begin{vmatrix} \frac{N(N+1)}{2} & 1 & 5 \\ \frac{N(N+1)(2N+1)}{6} & 2N+1 & 2N+1 \\ \frac{N^2(N+1)^2}{4} & 3N^2 & 3N \end{vmatrix} \\
 &= \frac{N(N+1)}{12} \begin{vmatrix} 6 & 1 & 5 \\ 2(2N+1) & 2N+1 & 2N+1 \\ 3N(N+1) & 3N^2 & 3N \end{vmatrix} \\
 &= \frac{N(N+1)}{12} \begin{vmatrix} 0 & 1 & 5 \\ 0 & 2N+1 & 2N+1 \\ 0 & 3N^2 & 3N \end{vmatrix} \quad C_1 \rightarrow C_1 - C_2 - C_3 \\
 &= 0
 \end{aligned}$$

36. If $a > 0$, $b > 0$, $c > 0$ are respectively the p^{th} , q^{th} , r^{th} terms of a G.P., then the value of the determinant

$$\begin{vmatrix} \log a & p & 1 \\ \log b & q & 1 \\ \log c & r & 1 \end{vmatrix} \text{ is}$$

(1) 1

(2) -1

(3) $\frac{abc}{pqr}$

(4) 0

Sol. Answer (4)

Let, $a = AR^{p-1}$, $b = AR^{q-1}$, $c = AR^{r-1}$

$$\begin{aligned}
 \therefore \begin{vmatrix} \log a & p & 1 \\ \log b & q & 1 \\ \log c & r & 1 \end{vmatrix} &= \begin{vmatrix} \log A + (p-1)\log R & p & 1 \\ \log A + (q-1)\log R & q & 1 \\ \log A + (r-1)\log R & r & 1 \end{vmatrix} \\
 &= \begin{vmatrix} (p-1)\log R & p & 1 \\ (q-1)\log R & q & 1 \\ (r-1)\log R & r & 1 \end{vmatrix} \quad C_1 \rightarrow C_1 - (\log A)C_3 \\
 &= \begin{vmatrix} 0 & p & 1 \\ 0 & q & 1 \\ 0 & r & 1 \end{vmatrix} \quad C_1 \rightarrow C_1 - (C_2 - C_3)\log R \\
 &= 0
 \end{aligned}$$

37. If a , b , c are positive and not equal then value of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ may be

(1) 1

(2) -3

(3) 2

(4) 4

Sol. Answer (2)

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc) \\
 = -(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) \\
 = -\frac{(a+b+c)}{2} \{(a-b)^2 + (b-c)^2 + (c-a)^2\} < 0$$

Hence possible value = -3

38. The greatest value of x satisfying the equation

$$\begin{vmatrix} 2^{11} - x & -2^{12} & 2^{11} \\ -2^{12} & 2^{11} - x & 2^{11} \\ 2^{11} & 2^{11} & -2^{12} - x \end{vmatrix} = 0 \text{ is}$$

(1) 3×2^{11}

(2) $2^{11} + 2^{13}$

(3) $2^{12} - 2^{11}$

(4) 5×2^{11}

Sol. Answer (1)Let $2^{11} = a$, then the given equation reduces to

$$\begin{vmatrix} a-x & -2a & a \\ -2a & a-x & a \\ a & a & -2a-x \end{vmatrix} = 0$$

Applying $C_1 \rightarrow C_1 + C_2 + C_3$

$$\begin{vmatrix} -x & -2a & a \\ -x & a-x & a \\ -x & a & -2a-x \end{vmatrix} = 0$$

$$\Rightarrow x \begin{vmatrix} 1 & -2a & a \\ 1 & a-x & a \\ 1 & a & -2a-x \end{vmatrix} = 0$$

$$\Rightarrow x \begin{vmatrix} 1 & -2a & a \\ 0 & 3a-x & 0 \\ 0 & 3a & -3a-x \end{vmatrix} = 0$$

$$\Rightarrow x[(3a-x)(-3a-x) - 0] = 0$$

$$\Rightarrow x(x-3a)(x+3a) = 0$$

 $\therefore$ Greatest value of x satisfying the given equation is $x = 3a = 3 \times 2^{11}$

39. The largest value of a third order determinant whose elements are equal to 1 or 0 is

(1) 0

(2) 2

(3) 4

(4) 6

Sol. Answer (2)

$$\begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 2$$

Option (2) is correct.

40. If $\begin{vmatrix} x^k & x^{k+2} & x^{k+3} \\ y^k & y^{k+2} & y^{k+3} \\ z^k & z^{k+2} & z^{k+3} \end{vmatrix} = (x-y)(y-z)(z-x) \left\{ \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right\}$, then

(1) $k = -2$

(2) $k = -1$

(3) $k = 0$

(4) $k = 1$

Sol. Answer (2)

$$\begin{vmatrix} x^k & x^{k+2} & x^{k+3} \\ y^k & y^{k+2} & y^{k+3} \\ z^k & z^{k+2} & z^{k+3} \end{vmatrix} = x^k \cdot y^k \cdot z^k (x-y)(y-z)(z-x) \{xy + yz + zx\}$$

$$= (xyz)^{k+1} (x-y)(y-z)(z-x) \left\{ \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right\}$$

$$\therefore k+1 = 0 \Rightarrow k = -1$$

Option (2) is correct.

41. If all elements of a third order determinant are equal to 1 or -1 , then determinant itself is

(1) An odd integer

(2) An even integer

(3) An imaginary number

(4) Multiple of 3

Sol. Answer (2)

Let third order determinant be

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

when $a_{ij} = 1$ or -1 then both $\frac{a_{21}}{a_{11}}$ and $\frac{a_{31}}{a_{11}}$ equals ± 1 .

Apply $R_2 \rightarrow R_2 - \left(\frac{a_{21}}{a_{11}}\right)R_1$ and $R_3 \rightarrow R_3 - \left(\frac{a_{31}}{a_{11}}\right)R_1$.

$$\therefore D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ 0 & 2\alpha & s\beta \\ 0 & 2\gamma & 2\delta \end{vmatrix} \text{ where } \alpha, \beta, \gamma, \delta \text{ are all } 1 \text{ or } -1.$$

Expand along $C_1 \Rightarrow \Delta = 4a_{11}\{\alpha\delta - \beta\gamma\}$.

which is an even integer hence option (2) is correct.

[Application of Determinants]

42. If A is a 3×3 matrix and $\det(3A) = k\{\det(A)\}$, then k is

(1) 9

(2) 6

(3) 1

(4) 27

Sol. Answer (4)

$$\det(3A) = k\{\det(A)\}$$

$$\Rightarrow 3^3 \det(A) = k\{\det(A)\}$$

$$\Rightarrow k = 27, \text{ option (4) is correct.}$$

43. If A is any of square matrix of order n , then $A(\text{adj}A)$ is equal to

- (1) 1 (2) $|A| \cdot I_n$ (3) 0 (4) $|A|^n$

Sol. Answer (2)

Clearly option (2) is correct.

44. If A is a square matrix of order 3, $|A| = 3$, then $|\text{adj} \text{adj} A|$ is equal to

- (1) 3^5 (2) 3^7 (3) 9 (4) 81

Sol. Answer (4)

$$|\text{adj}A| = 3^2 = 9$$

$$|\text{adj}(\text{adj}A)| = 9^2 = 81$$

45. If $A = \begin{bmatrix} 11 & 7 \\ -13 & 17 \end{bmatrix}$, then $\text{adj}(\text{adj} A)$ is

- (1) $\begin{bmatrix} 17 & -7 \\ 13 & 11 \end{bmatrix}$ (2) $\begin{bmatrix} 11 & 7 \\ -13 & 17 \end{bmatrix}$ (3) $\begin{bmatrix} -17 & 7 \\ 13 & -11 \end{bmatrix}$ (4) $\begin{bmatrix} -11 & 7 \\ -13 & 17 \end{bmatrix}$

Sol. Answer (2)

For a square matrix of order 2×2 , $\text{adj}(\text{adj} A) = A$

46. If $A = \begin{bmatrix} \sin \alpha & -\cos \alpha & 0 \\ \cos \alpha & \sin \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then A^{-1} is equal to

- (1) $-A^T$ (2) A (3) $\text{adj} A$ (4) A^2

Sol. Answer (3)

$$|A| = \sin^2 \alpha + \cos^2 \alpha = 1 (\neq 0).$$

Hence A^{-1} exists. Since, $A^{-1} = \frac{1}{|A|} \text{adj}(A)$ and for the given question $|A| = 1$, hence $A^{-1} = \text{adj}(A)$

47. If A is a matrix of order 3 and $|A| = 2$, then $|\text{adj} A|$ is

- (1) 1 (2) 2 (3) 8 (4) 4

Sol. Answer (4)

$\det(\text{adj} A) = (\det A)^{n-1}$, where n is the order of matrix A , therefore $|\text{adj} A| = |A|^{3-1} = 2^2 = 4$

48. Let $A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$. The only correct statement about the matrix A is

- (1) A is a zero matrix (2) $A = (-1)I_3$ (3) A^{-1} doesn't exist (4) $A^2 = I$

Sol. Answer (4)

Clearly, options (1), (2) & (3) are false. Option (4) can be easily verified as

$$\begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

[Solutions of Equations]

49. If the lines $ax + by + 1 = 0$, $x + by + 1 = 0$ and $x + y + c = 0$ (a, b, c being distinct and different from unity are concurrent), then the value of $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is

- (1) 0 (2) 1 (3) -1 (4) 2

Sol. Answer (2)

$$\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$$

$$R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_3$$

$$\Rightarrow \begin{vmatrix} a-1 & 1-b & 0 \\ 0 & b-1 & 1-c \\ 1 & 1 & c \end{vmatrix} = 0$$

$$\Rightarrow (a-1)(b-1)c - (a-1)(1-c) + (1-b)(1-c) = 0$$

Divided by $(1-a)(1-b)(1-c)$

$$\Rightarrow \frac{c}{1-c} + \frac{1}{1-b} + \frac{1}{1-a} = 0$$

$$\Rightarrow \frac{1+c-1}{1-c} + \frac{1}{1-b} + \frac{1}{1-a} = 0$$

$$\Rightarrow \frac{1}{1-c} + \frac{1}{1-b} + \frac{1}{1-a} = 1$$

50. If the system of equations

$$x + 4ay + az = 0$$

$$x + 3by + bz = 0$$

$$x + 2cy + cz = 0$$

have a non-zero solution, then a, b, c ($\neq 0$) are in

- (1) AP (2) GP (3) HP (4) AGP

Sol. Answer (3)

$$\begin{vmatrix} 1 & 4a & a \\ 1 & 3b & b \\ 1 & 2c & c \end{vmatrix} = 0$$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$$

$$\begin{vmatrix} 1 & 4a & a \\ 0 & 3b-4a & b-a \\ 0 & 2c-4a & c-a \end{vmatrix} = 0$$

$$\Rightarrow ab + bc = 2ac$$

$$\Rightarrow \frac{2}{b} = \frac{1}{a} + \frac{1}{c}$$

$$\Rightarrow a, b, c \text{ are in HP}$$

51. Let $x + y + z = 6$, $4x + \lambda y - \lambda z = 0$, $3x + 2y - 4z = -5$. The value of λ for which given system of equations does not have a unique solution is

(1) 0 (2) 1 (3) 2 (4) 3

Sol. Answer (4)

To not have unique solution,

$$\begin{vmatrix} 1 & 1 & 1 \\ 4 & \lambda & -\lambda \\ 3 & 2 & -4 \end{vmatrix} = 0$$

$$1(2\lambda - 0) - 1(\lambda - 3) + 1(2 - 2) = 0$$

$$\lambda - 3 = 0$$

$$\lambda = 3$$

52. If the system of the equations $x - ky - z = 0$, $kx - y - z = 0$, $x + y - z = 0$ has a non-zero solution, then the possible value of k are

(1) $-1, 2$ (2) $1, 2$ (3) $0, 1$ (4) $-1, 1$

Sol. Answer (4)

To have non-zero solution,

$$\begin{vmatrix} 1 & -k & -1 \\ k & -1 & -1 \\ 1 & 1 & -1 \end{vmatrix} = 0$$

$$1(1 + 1) + k(-k + 1) - 1(k + 1) = 0$$

$$2 - k^2 + k - k - 1 = 0$$

$$-k^2 + 1 = 0$$

$$k = \pm 1$$

53. The system of equations $ax + 4y + z = 0$, $bx + 3y + z = 0$, $cx + 2y + z = 0$ has non-trivial solution if a, b, c are in

(1) AP (2) GP (3) HP (4) None of these

Sol. Answer (1)

For non-trivial solutions,

$$\begin{vmatrix} a & 4 & 1 \\ b & 3 & 1 \\ c & 2 & 1 \end{vmatrix} = 0 \Rightarrow a - 2b + c = 0$$

$\therefore a, b, c$ are in A.P.

54. The number of 3×3 matrices A whose entries are either 0 or 1 and for which the system $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has exactly

two distinct solutions, is

[IIT-JEE 2010]

(1) 0 (2) $2^9 - 1$ (3) 168 (4) 2

Sol. Answer (1)

The equation $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has two distinct solutions. It should be noted here that the given equation is linear

equation in 3 variables, which may have no solution, or unique solution or infinitely many solutions.

Hence there does not exist any matrix A such that the given equation has exactly two solutions and consequently number of 3×3 matrices is 0.

[Miscellaneous]

55. Let $\omega \neq 1$ be a cube root of unity and S be the set of all non-singular matrices of the form

$$\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$$

where each of a, b and c is either ω or ω^2 . Then the number of distinct matrices in the set S is **[IIT-JEE 2011]**

(1) 2

(2) 6

(3) 4

(4) 8

Sol. Answer (1)

We have,

$$M = \begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$$

$$\Rightarrow |M| = \begin{vmatrix} 0 & a - \omega^2 & b + c + 1 \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{vmatrix}$$

$$= -\omega(a - \omega^2 - b\omega - c\omega - \omega) + \omega^2(ac - \omega^2c - b - c - 1)$$

$$= -(a + c)\omega + ac\omega^2 + 1$$

$$a + c \neq 1, ac \neq 1$$

Since a, b, c are ω or ω^2

$$\Rightarrow a = c$$

$$\text{If } a = \omega \Rightarrow c = \omega$$

$$\therefore \text{Number of ways of selecting } a, b, c = 1 \times 1 \times 2 = 2$$

$$\text{If } a = \omega^2, \text{ then number of ways} = 1 \times 1 \times 1 = 2$$

Total number of distinct matrices in the given set $S = 4$.

56. Let $P = [a_{ij}]$ be a 3×3 matrix and let $Q = [b_{ij}]$, where $b_{ij} = 2^{i+j}a_{ij}$ for $1 \leq i, j \leq 3$. If the determinant of P is 2, then the determinant of the matrix Q is **[IIT-JEE 2012]**

(1) 2^{10} (2) 2^{11} (3) 2^{12} (4) 2^{13} **Sol.** Answer (4)

$$\text{Given, } P = [a_{ij}]_{3 \times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Now, $Q = 2^{i+j} a_{ij} = [b_{ij}]$

$$\Rightarrow Q = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} = \begin{bmatrix} 2^2 a_{11} & 2^3 a_{12} & 2^4 a_{13} \\ 2^3 a_{21} & 2^4 a_{22} & 2^5 a_{23} \\ 2^4 a_{31} & 2^5 a_{32} & 2^6 a_{33} \end{bmatrix}$$

$$\text{So, } |Q| = 2^{12} \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 2^{13}.$$

(as $|P| = 2$)

SECTION - B

Objective Type Questions (More than one options are correct)

1. If $f(x)$ and $g(x)$ are functions such that $f(x+y) = f(x)g(y) + g(x)f(y)$, then

$$\begin{vmatrix} f(\alpha) & g(\alpha) & f(\alpha+\theta) \\ f(\beta) & g(\beta) & f(\beta+\theta) \\ f(\gamma) & g(\gamma) & f(\gamma+\theta) \end{vmatrix} \text{ is independent of}$$

(1) α

(2) β

(3) γ

(4) θ

Sol. Answer (1, 2, 3, 4)

$$\begin{vmatrix} f(\alpha) & g(\alpha) & f(\alpha+\theta) \\ f(\beta) & g(\beta) & f(\beta+\theta) \\ f(\gamma) & g(\gamma) & f(\gamma+\theta) \end{vmatrix}$$

$$= \begin{vmatrix} f(\alpha) & g(\alpha) & f(\alpha)g(\theta) + g(\alpha)f(\theta) \\ f(\beta) & g(\beta) & f(\beta)g(\theta) + g(\beta)f(\theta) \\ f(\gamma) & g(\gamma) & f(\gamma)g(\theta) + g(\gamma)f(\theta) \end{vmatrix}$$

$$= \begin{vmatrix} f(\alpha) & g(\alpha) & 0 \\ f(\beta) & g(\beta) & 0 \\ f(\gamma) & g(\gamma) & 0 \end{vmatrix} \quad C_3 \rightarrow C_3 - g(\theta)C_1 - f(\theta)C_2$$

$$= 0$$

Hence, independent of α, β, γ and θ .

2. The digits A, B, C are such that the three digit numbers $A88, 6B8, 86C$ are divisible by 72, then the determinant

$$\begin{vmatrix} A & 6 & 8 \\ 8 & B & 6 \\ 8 & 8 & C \end{vmatrix} \text{ is divisible by}$$

(1) 72

(2) 144

(3) 288

(4) 1216

Sol. Answer (1, 2, 3)

3 digit numbers divisible by 72, include 288, 648 and 864.

Hence, the only possibility is

$$A = 2, B = 4, C = 4$$

$$\therefore \Delta = \begin{vmatrix} 2 & 6 & 8 \\ 8 & 4 & 6 \\ 8 & 8 & 4 \end{vmatrix} = 288$$

$\therefore \Delta$ is divisible by 72, 144 and 288.

3. If $\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix} = 0$ then

(1) a, b, c are in A.P.

(2) a, b, c are in G.P.

(3) α is a root of $ax^2 + 2bx + c = 0$

(4) $(x - \alpha)$ is a factor of $ax^2 + 2bx + c = 0$

Sol. Answer (2, 3, 4)

$$\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix} = 0$$

$$\Rightarrow R_3 \rightarrow R_3 - (\alpha R_1 + R_2)$$

$$\begin{vmatrix} a & b & 0 \\ b & c & 0 \\ a\alpha + b & b\alpha + c & -(a\alpha^2 + b\alpha + b\alpha + c) \end{vmatrix} = 0$$

$$\Rightarrow -(a\alpha^2 + 2b\alpha + c)(b^2 - ac) = 0$$

$$b^2 = ac, \quad a\alpha^2 + 2b\alpha + c = 0$$

Hence option (2, 3, 4) are true

4. The value of the determinant $\begin{vmatrix} 1 & bc & a(b+c) \\ 1 & ca & b(a+c) \\ 1 & ab & c(a+b) \end{vmatrix}$ doesn't depend on

(1) a

(2) b

(3) c

(4) $a + b + c$

Sol. Answer (1, 2, 3, 4)

$$\Delta = \begin{vmatrix} 1 & bc & a(b+c) \\ 1 & ca & b(a+c) \\ 1 & ab & c(a+b) \end{vmatrix}$$

$$= \begin{vmatrix} 0 & c(b-a) & c(a-b) \\ 0 & a(c-b) & a(b-c) \\ 1 & ab & c(a+b) \end{vmatrix} \quad (R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_1 - R_3)$$

$$\therefore \Delta = \begin{vmatrix} c(b-a) & c(a-b) \\ a(c-b) & a(b-c) \end{vmatrix} = ac(a-b)(b-c) \begin{vmatrix} -1 & 1 \\ -1 & 1 \end{vmatrix} = 0$$

$\therefore \Delta$ does not depend on a, b, c .

5. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then

(1) $A^3 - 6A^2 + 11A - 6I = 0$

(2) $A^{-1} = \frac{1}{6}(A^2 - 6A + 11I)$

(3) A^2 is non-singular

(4) A is singular

Sol. Answer (1, 2, 3)

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & -2 & 4-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda) \{(1-\lambda)(4-\lambda) + 2\} = 0$$

$$\Rightarrow (1-\lambda)(\lambda^2 - 5\lambda + 4 + 2) = 0$$

$$(\lambda - 1)(\lambda^2 - 5\lambda + 6) = 0$$

$$\lambda^3 - 5\lambda^2 + 6\lambda + 5\lambda - \lambda^2 - 6 = 0$$

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

Hence the characteristics equation of the matrix is

$$A^3 - 6A^2 + 11A - 6I = 0$$

$$\Rightarrow A^{-1} = \frac{1}{6}(A^2 - 6A + 11I)$$

$$|A^2| \neq 0$$

$$|A| \neq 0$$

Hence options (1, 2, 3) are correct

6. A is a matrix of order 3×3 . If $A' = A$ and five entries in the matrix are of one kind and remaining four are of another kind, then the maximum number of such matrices is greater than or equal to

(1) 9

(2) 10

(3) 11

(4) 8

Sol. Answer (1, 2, 3, 4)

Since, the matrix is symmetric, it can be of the following type

$$A = \begin{bmatrix} a & x & y \\ x & b & z \\ y & z & c \end{bmatrix}$$

Case (i) : Let $a = b = c = x \neq y = z$

In this case total matrices $= {}^3C_1 = 3$

Case (ii) : Let $a = x = y \neq b = c = z$

In this case total matrices $= {}^3C_2 \times {}^3C_1 = 9$

Hence, maximum number of such matrices $= 3 + 9 = 12$

7. The values of α for which the system of equations $x + y + z = 1$, $x + 2y + 4z = \alpha$, $x + 4y + 10z = \alpha^2$ is consistent, are given by

- (1) 1, 2 (2) -1, 2 (3) 1, -2 (4) -1, -2

Sol. Answer (1)

Clearly, the determinant

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 3 \\ 1 & 3 & 9 \end{vmatrix} = 0 \quad (C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1)$$

therefore, the given equations are consistent, if each of the three determinants

$$\begin{vmatrix} 1 & 1 & 1 \\ \alpha & 2 & 4 \\ \alpha^2 & 4 & 10 \end{vmatrix}, \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & 4 \\ 1 & \alpha^2 & 10 \end{vmatrix}, \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & \alpha \\ 1 & 4 & \alpha^2 \end{vmatrix} \text{ are zero.}$$

Solving, $\alpha = 1, 2$

8. The system of equations

$$x - y \cos \theta + z \cos 2\theta = 0$$

$$-x \cos \theta + y - z \cos \theta = 0$$

$$x \cos 2\theta - y \cos \theta + z = 0$$

has non-trivial solution for θ equals

- (1) $\frac{\pi}{3}$ (2) $\frac{\pi}{6}$ (3) $\frac{2\pi}{3}$ (4) $\frac{\pi}{12}$

Sol. Answer (1, 2, 3, 4)

For non-trivial solution

$$\begin{vmatrix} 1 & -\cos \theta & \cos 2\theta \\ -\cos \theta & 1 & -\cos \theta \\ \cos 2\theta & -\cos \theta & 1 \end{vmatrix} = 0$$

on expansion, the determinant equals

$$\begin{aligned} & (1 - \cos^2 \theta) + \cos \theta (-\cos \theta + \cos \theta \cos 2\theta) + \cos 2\theta (\cos^2 \theta - \cos 2\theta) \\ &= 1 - 2\cos^2 \theta + 2\cos^2 \theta \cos 2\theta - \cos^2 2\theta \\ &= 1 - 2\cos^2 \theta (1 - \cos 2\theta) - \cos^2 2\theta \\ &= 1 - 4\cos^2 \theta \sin^2 \theta - \cos^2 2\theta \\ &= 1 - 4\cos^2 \theta (1 - \cos^2 \theta) - (2\cos^2 \theta - 1)^2 \\ &= 1 - (4\cos^2 \theta - 4\cos^4 \theta) - (4\cos^4 \theta - 4\cos^2 \theta + 1) \\ &= 0, \text{ for all values of } \theta. \end{aligned}$$

9. If the adjoint of a 3×3 matrix P is $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$, then the possible value(s) of the determinant of P is (are)

[IIT-JEE 2012]

- (1) -2 (2) -1 (3) 1 (4) 2

Sol. Answer (1, 4)

$$\text{adj } P = \begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$$

$$|\text{adj } P| = 1(3 - 7) - 4(6 - 7) + 4(2 - 1) = -4 + 4 + 4 = 4$$

$$\therefore |\text{adj } P| = |P|^2$$

$$\Rightarrow |P|^2 = 4$$

$$\Rightarrow |P| = \pm 2$$

10. For 3×3 matrices M and N , which of the following statement(s) is (are) **NOT** correct? [JEE(Advanced) 2013]

- (1) $N^T M N$ is symmetric or skew symmetric, according as M is symmetric or skew symmetric
 (2) $MN - NM$ is skew symmetric for all symmetric matrices M and N
 (3) MN is symmetric for all symmetric matrices M and N
 (4) $(\text{adj } M)(\text{adj } N) = \text{adj}(MN)$ for all invertible matrices M and N

Sol. Answer (3, 4)

$$(N^T M N)^T = -N^T M^T (N^T)^T$$

$$N^T M^T N$$

- (1) If M is skew symmetric, $(N^T M N)^T = -N^T M N$,

Hence skew symmetric.

$$\text{If } M \text{ is symmetric, } (M^T M N)^T = N^T M N,$$

Hence symmetric.

$\therefore$ Option (1) is correct.

- (2) $(MN - NM)^T = (MN)^T - (NM)^T = N^T M^T - M^T N^T = -(M^T M^T - N^T M^T) = -(MN - NM)$

$\therefore$ Skew symmetric option (2) is correct.

- (3) $(MN)^T = N^T M^T$

Symmetry and skew symmetry depend on nature of M and N , option (3) is incorrect.

- (4) $\text{adj}(MN) = \text{adj}(N) \text{adj } M$

Option (4) is incorrect.

11. Let M and N be two 3×3 matrices such that $MN = NM$. Further, if $M \neq N^2$ and $M^2 = N^4$, then

[JEE(Advanced)-2014]

- (1) Determinant of $(M^2 + MN^2)$ is 0
 (2) There is a 3×3 non-zero matrix U such that $(M^2 + MN^2)U$ is the zero matrix
 (3) Determinant of $(M^2 + MN^2) \geq 1$
 (4) For a 3×3 matrix U , if $(M^2 + MN^2)U$ equals the zero matrix then U is the zero matrix

Sol. Answer (1, 2)

$$M \neq N^2 \Rightarrow M - N^2 \neq 0$$

$$M^2 - N^4 = 0 \Rightarrow (M - N^2)(M + N^2) + N^2M - MN^2 = 0$$

$$\text{as } MN = NM$$

$$\Rightarrow MN^2 = NMN$$

$$= NNM$$

$$= N^2M$$

$$\text{So, } (M - N^2)(M + N^2) = 0$$

$$\text{So either } M + N^2 = 0 \quad \text{or } M - N^2 \text{ and } M + N^2 \text{ both are singular.}$$

$$\text{So, there exist a } 3 \times 3 \text{ non-zero matrix } U \text{ i.e., } M - N^2 \text{ such that}$$

$$(M + N^2)U = 0 \Rightarrow (M^2 + MN^2)U = 0$$

$$\text{Also, } |M^2 + MN^2| = |M||M + N^2| = 0$$

12. Which of the following values of α satisfy the equation

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\alpha ?$$

[JEE(Advanced)-2015]

(1) -4

(2) 9

(3) -9

(4) 4

Sol. Answer (2, 3)

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} \begin{matrix} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{matrix} \text{ (Take common)}$$

$$2 \begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ 3+2\alpha & (3+4\alpha) & (3+6\alpha) \\ (4+2\alpha) & (4+4\alpha) & (4+6\alpha) \end{vmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$2 \begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ 3+2\alpha & 3+4\alpha & 3+6\alpha \\ 1 & 1 & 1 \end{vmatrix}$$

$$\begin{matrix} C_2 \rightarrow C_2 - C_1 \\ C_3 \rightarrow C_3 - C_1 \end{matrix} \text{ (Take common)}$$

$$\therefore 2\alpha \cdot 2\alpha \begin{vmatrix} (1+\alpha)^2 & (2+3\alpha) & (2+4\alpha) \\ 3+2\alpha & 2 & 2 \\ 1 & 0 & 0 \end{vmatrix}$$

$$4\alpha^2 |-2\alpha| = -8\alpha^3 = -648\alpha$$

$$\alpha = 0, \alpha^2 = 81, \alpha = \pm 9$$

13. Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$. Suppose $Q = [q_{ij}]$ is a matrix such that $PQ = kI$, where $k \in \mathbb{R}$, $k \neq 0$ and I is

the identity matrix of order 3. If $q_{23} = -\frac{k}{8}$ and $\det(Q) = \frac{k^2}{2}$, then

[JEE(Advanced)-2016]

- (1) $\alpha = 0, k = 8$ (2) $4\alpha - k + 8 = 0$ (3) $\det(P \operatorname{adj}(Q)) = 2^9$ (4) $\det(Q \operatorname{adj}(P)) = 2^{13}$

Sol. Answer (2, 3)

$$PQ = kI$$

$$\Rightarrow |P| |Q| = k^3$$

$$\Rightarrow |P| \frac{k^2}{2} = k^3$$

$$\Rightarrow |P| = 2k = 12\alpha + 20 \quad \dots(i)$$

$$\text{Also, } Q = kP^{-1}$$

$$= k \cdot \frac{\operatorname{adj}P}{|P|} = \frac{\operatorname{adj}P}{2}$$

$$\text{So, } q_{23} = -\left(\frac{3\alpha + 4}{2}\right) = \frac{-k}{8}$$

$$\Rightarrow 3\alpha + 4 = \frac{k}{4} \quad \dots(ii)$$

On solving (i) & (ii),

$$k = 4, \alpha = -1, |P| = 8, |Q| = 8$$

$$\Rightarrow 4\alpha - k + 8 = 0$$

$$|P \operatorname{adj}Q| = |P| |Q|^2 = 8^3 = 2^9$$

$$|Q \operatorname{adj}P| = |Q| |P|^2 = 8^3 = 2^9$$

14. Let $a, \lambda, \mu \in \mathbb{R}$. Consider the system of linear equations

$$ax + 2y = \lambda$$

$$3x - 2y = \mu$$

Which of the following statement(s) is(are) correct?

[JEE(Advanced)-2016]

- (1) If $a = -3$, then the system has infinitely many solutions for all values of λ and μ
 (2) If $a \neq -3$, then the system has a unique solution for all values of λ and μ
 (3) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$
 (4) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$

Sol. Answer (2, 3, 4)

$$ax + 2y = \lambda$$

$$3x - 2y = \mu$$

$$\frac{a}{3} \neq \frac{2}{-2} \text{ for unique solution}$$

$$a \neq -3$$

$$\frac{a}{3} = \frac{2}{-2} = \frac{\lambda}{\mu} \text{ for infinite solution}$$

$$a = -3, \lambda + \mu = 0$$

$$\frac{\mu}{\lambda} \neq -1$$

For no solution,

$$a = -3 \text{ but } \mu + \lambda \neq 0$$

SECTION - C

Linked Comprehension Type Questions

Comprehension-I

Consider a matrix $A = [a_{ij}]_{n \times n}$. Form the matrix $A - \lambda I$, λ being a number, real or complex.

$$A - \lambda I = \begin{bmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} - \lambda & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} - \lambda \end{bmatrix}$$

Then $\det(A - \lambda I) = (-1)^n [\lambda^n + b_1 \lambda^{n-1} + b_2 \lambda^{n-2} + \dots + b_n]$.

An important theorem tells us that the matrix A satisfies the equation $x^n + b_1 x^{n-1} + b_2 x^{n-2} + \dots + b_n = 0$. This equation

is called the characteristic equation of A . For all the questions on this passage, take $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

1. The matrix A satisfies the matrix equation

$$(1) A^2 - 4A - 5I = 0$$

$$(2) A^2 - 4A - 5 = 0$$

$$(3) A^2 + 4A - 5I = 0$$

$$(4) A^2 + 4A - 5 = 0$$

Sol. Answer (1)

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix} = 0 \Rightarrow (1-\lambda)(3-\lambda) - 8 = 0 \Rightarrow \lambda^2 - 4\lambda - 5 = 0 \Rightarrow A^2 - 4A - 5I = 0$$

2. Which of the following is inverse of A ?

$$(1) \frac{1}{5}(A - 4I)$$

$$(2) \frac{1}{5}(A + 4I)$$

$$(3) \frac{1}{4}(A - 5I)$$

$$(4) \frac{1}{4}(A + 5I)$$

Sol. Answer (1)

$$A^2 - 4A - 5I = 0$$

$$\Rightarrow A^{-1}(A^2 - 4A - 5I) = 0$$

$$\Rightarrow A - 4I - 5A^{-1} = 0$$

$$\Rightarrow 5A^{-1} = A - 4I$$

$$\Rightarrow A^{-1} = \frac{1}{5}(A - 4I)$$

3. The matrix $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$, when expressed as a linear polynomial in A , is

- (1) $A + 5I$ (2) $A - 5I$ (3) $-A + 5I$ (4) $-A - 5I$

Sol. Answer (1)

$$\begin{aligned} & A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I \\ &= A^3(A^2 - 4A - 5) - 2A^3 + 11A^2 - A - 10I \\ &= -2A(A^2 - 4A - 5I) + 3A^2 - 11A - 10I \\ &= 3(4A + 5I) - 11A - 10I \\ &= A + 5I \end{aligned}$$

Comprehension-II

Matrix theory can be applied to investigate the conditions under which a given system of linear equations possesses unique, infinite or no solutions. Consider the system of 3 non-homogeneous linear equations in 3 unknowns

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \rho$$

and answer the questions that follow.

1. The system possesses a unique solution if

- (1) $\lambda \neq 3$ (2) $\lambda \neq 3, \rho \neq 10$ (3) $\rho \neq 10$ (4) $\rho = 10$

Sol. Answer (1)

For unique solution,

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{vmatrix} \neq 0$$

$$\Rightarrow \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & \lambda - 2 \end{vmatrix} \neq 0$$

$$(C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_2)$$

$$\Rightarrow (\lambda - 2) - 1 \neq 0$$

$$\Rightarrow \lambda \neq 3$$

2. The system possesses no solutions if

- (1) $\lambda \neq 3, \rho = 10$ (2) $\lambda = 3, \rho \neq 10$ (3) $\rho = 10$ (4) $\rho = 3, \lambda \neq 10$

Sol. Answer (2)

For no solution

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{vmatrix} = 0 \text{ and } \rho \neq 10$$

$$\text{i.e., } \lambda = 3 \text{ and } \rho \neq 10.$$

The last two equations must not be identical.

3. The system possesses infinite solutions if

(1) $\lambda = 3, \rho \neq 10$

(2) $\lambda \neq 3, \rho = 10$

(3) $\lambda = 3, \rho = 10$

(4) $\lambda \neq 3, \rho \neq 10$

Sol. Answer (3)

For infinite solutions

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{vmatrix} = 0 \text{ and } \rho = 10$$

i.e., $\lambda = 3$ and $\rho = 10$

The last two equations must be identical.

Comprehension-III

A and B are two matrices of same order 3×3 , where

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 5 & 6 & 8 \end{bmatrix}, B = \begin{bmatrix} 3 & 2 & 5 \\ 2 & 3 & 8 \\ 7 & 2 & 9 \end{bmatrix}$$

1. The value of $\text{adj}(\text{adj } A)$ is

(1) $-A$

(2) $4A$

(3) $8A$

(4) $16A$

Sol. Answer (1)

2. The value of $|\text{adj}(AB)|$ is

(1) 24

(2) 24^2

(3) 24^3

(4) 65

Sol. Answer (2)

3. Value of $|\text{adj}(\text{adj}(\text{adj}(\text{adj } A)))|$ is

(1) 2^4

(2) 2^9

(3) 1

(4) 2^{19}

Sol. Answer (3)

Comprehension-IV

Let $\mathcal{A}$ be the set of all 3×3 symmetric matrices all of whose entries are either 0 or 1. Five of these entries are 1 and four of them are 0. **[IIT-JEE 2009]**

1. The number of matrices in $\mathcal{A}$ is

(1) 12

(2) 6

(3) 9

(4) 3

Sol. Answer (1)

The matrix is of order 3×3 where there are 9 elements. This has to be symmetric matrix with five of the entries as 1 and four of the entries as zero. For the matrix to be symmetric (i) either all the three elements along principal diagonal should be (1) or exactly one element along principal diagonal should be 1.

The possible matrixes are as follows

$$A_1 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, A_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, A_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$B_1 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, B_2 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, B_3 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$C_1 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, C_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}, C_3 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$D_1 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}, D_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, D_3 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

There are 12 such matrices.

2. The number of matrices A in $\mathbb{R}$ for which the system of linear equations $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has a unique solution, is

- (1) Less than 4 (2) At least 4 but less than 7
(3) At least 7 but less than 10 (4) At least 10

Sol. Answer (2)

The determinant corresponding to matrices B_2, B_3, C_1, C_3, D_1 and D_2 are non-zero. Therefore in these three cases, the given linear equations will have unique solution. Number of required matrices in this case is 6.

3. The number of matrices A in $\mathbb{R}$ for which the system of linear equations $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is inconsistent, is

- (1) 0 (2) More than 2 (3) 2 (4) 1

Sol. Answer (2)

(i) If we take (i) $A_1 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then $|A_1| = 0$

$$\Delta x = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \neq 0, \text{ hence in this case system is inconsistent.}$$

(ii) If we take $A_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \Rightarrow |A_2| = 0$

$$\Delta x = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \neq 0, \text{ hence system is inconsistent.}$$

(iii) If we take $A_3 = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{vmatrix} \Rightarrow |A_3| = 0$

$\Delta x = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{vmatrix} = 0$, Similarly $\Delta y = A_3 = 0$. In this case we get infinite solution.

(iv) If we take $B_1 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{vmatrix} \Rightarrow |B_1| = 0$. In this case $\Delta x = \Delta y = \Delta z = 0$

Hence infinite solution consistent system.

Comprehension-V

Let a, b and c be three real numbers satisfying

$$[a \ b \ c] \begin{bmatrix} 1 & 9 & 7 \\ 8 & 2 & 7 \\ 7 & 3 & 7 \end{bmatrix} = [0 \ 0 \ 0] \dots (E)$$

[IIT-JEE 2011]

1. If the point $P(a, b, c)$, with reference to (E), lies on the plane $2x + y + z = 1$, then the value of $7a + b + c$ is

(1) 0 (2) 12 (3) 7 (4) 6

Sol. Answer (4)

From the given condition,

$$a + 8b + 7c = 0$$

$$9a + 2b + 3c = 0$$

$$7a + 7b + 7c = 0$$

$$a + b + c = 0$$

Also, $2a + b + c = 1$

$$\Rightarrow a = 1$$

$$\Rightarrow 7a + b + c = 6a + (a + b + c) = 6$$

2. Let ω be a solution of $x^3 - 1 = 0$ with $\text{Im}(\omega) > 0$. If $a = 2$ with b and c satisfying (E), then the value of $\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c}$ is equal to

(1) -2 (2) 2 (3) 3 (4) -3

Sol. Answer (1)

When $a = 2 \Rightarrow b + c = -2$

Also, $8b + 7c = -2$

$$\Rightarrow b = 12 \text{ and } c = -14$$

$$\begin{aligned} \text{Thus, } \frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c} &= \frac{3}{\omega^2} + \frac{1}{\omega^{12}} + \frac{3}{\omega^{-14}} \\ &= 1 + 3\omega + 3\omega^2 \\ &= 1 - 3 \\ &= -2 \end{aligned}$$

3. Let $b = 6$, with a and c satisfying (E). If α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then

$$\sum_{n=0}^{\infty} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)^n$$

(1) 6

(2) 7

(3) $\frac{6}{7}$ (4) ∞ **Sol.** Answer (2)

$$\text{For } b = 6 \Rightarrow a + c = -6$$

$$a + 7c = -48$$

$$a + c = -6$$

$$\text{when } a = 1, c = -7$$

α and β are roots of the given equation $ax^2 + bx + c = 0$, hence $\alpha + \beta = -6$, $\alpha\beta = -7$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{6}{7}$$

$$\sum_{n=0}^{\infty} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)^n = 1 + \frac{6}{7} + \left(\frac{6}{7} \right)^2 + \dots = \frac{1}{1 - \frac{6}{7}} = 7$$

SECTION - D

Assertion-Reason Type Questions

1. Consider the matrix $A = \begin{bmatrix} 0 & -h & -g \\ h & 0 & -f \\ g & f & 0 \end{bmatrix}$

STATEMENT-1 : $\det A = 0$.

and

STATEMENT-2 : The value of the determinant of a skew symmetric matrix of odd order is always zero.

Sol. Answer (1)

$$|A| = \begin{vmatrix} 0 & -h & -g \\ h & 0 & -f \\ g & f & 0 \end{vmatrix} = 0 + h \begin{vmatrix} h & -f \\ g & 0 \end{vmatrix} - g \begin{vmatrix} h & 0 \\ g & f \end{vmatrix} = 0 + h(0 + gf) - g(hf - 0) = 0$$

2. Consider the determinants $\Delta = \begin{vmatrix} 2 & -1 & 3 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix}$, $\Delta' = \begin{vmatrix} 2 & 0 & -2 \\ -2 & -1 & 1 \\ -4 & 1 & 3 \end{vmatrix}$

STATEMENT-1 : $\Delta' = \Delta^2$.

and

STATEMENT-2 : The determinant formed by the cofactors of the elements of a determinant of order 3 is equal in value to the square of the original determinant.

Sol. Answer (1)

Clearly, Δ' is formed by the cofactor of Δ

$$\therefore \Delta' = \Delta^2$$

3. STATEMENT-1 : Matrix $\begin{bmatrix} a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \end{bmatrix}$ is a diagonal matrix.

and

STATEMENT-2 : $A = [a_{ij}]$ is a square matrix such that $a_{ij} = 0 \forall i \neq j$, then A is called diagonal matrix.

Sol. Answer (4)

A is not diagonal matrix

$\therefore A$ is not square matrix

Option (4) is correct.

4. STATEMENT-1 : Inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 7 \end{bmatrix}$ does not exist.

and

STATEMENT-2 : If any matrix is singular, then its inverse does not exist.

Sol. Answer (1)

$\therefore |A| = 0$, A^{-1} does not exist.

Option (1) correct.

5. STATEMENT-1 : The system of equations $x + ky + 3z = 0$, $3x + ky - 2z = 0$, $2x + 3y - 4z = 0$, possesses a non-trivial solution, then value of k is $\frac{31}{2}$.

and

STATEMENT-2 : Three linear equations in x, y, z can never, have exactly two solutions.

Sol. Answer (4)

$$\therefore \Delta = \begin{vmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{vmatrix} = 0 \Rightarrow 20k + 33 - 22k = 0$$

$$\Rightarrow k = \frac{33}{2}$$

Option (4) is correct.

6. STATEMENT-1 : $f(x) = \begin{vmatrix} (1+x)^{21} & (1+x)^{22} & (1+x)^{23} \\ (1+x)^{31} & (1+x)^{32} & (1+x)^{33} \\ (1+x)^{41} & (1+x)^{42} & (1+x)^{43} \end{vmatrix}$, then coefficient of x in $f(x)$ is zero.

and

STATEMENT-2 : If $F(x) = A_0 + A_1x + \dots + A_nx^n$, then $A_1 = F'(0)$, dash denotes differential coefficient.

Sol. Answer (1)

$$f(x) = a_1 + 2a_2x +$$

$$\text{or } f(0) = a_1$$

$$\therefore a_1 = \begin{vmatrix} 21 & 22 & 23 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ 31 & 32 & 33 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 41 & 42 & 43 \end{vmatrix} = 0$$

Option (1) is correct.

7. Consider the system of equations

$$x - 2y + 3z = -1$$

$$-x + y - 2z = k$$

$$x - 3y + 4z = 1$$

STATEMENT-1 : The system of equations has no solution for $k \neq 3$.

and

STATEMENT-2 : The determinant $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$, for $k \neq 3$.

[IIT-JEE 2008]

Sol. Answer (1)

The given system of equations can be expressed as

$$\begin{bmatrix} 1 & -2 & 3 \\ 1 & -3 & 4 \\ -1 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ k \end{bmatrix}$$

Applying $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_1 + R_3$

$$\sim \begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ k-1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ k-3 \end{bmatrix} \text{ by } R_3 \rightarrow R_3 - R_2$$

When $k \neq 3$, the given system of equations has no solution.

$\Rightarrow$ Statement 1 is true. Clearly statement-2 is also true as it is rearrangement of rows and columns of

$$\begin{bmatrix} 1 & -2 & 3 \\ 1 & -3 & 4 \\ -1 & 1 & -2 \end{bmatrix}$$

Hence, option (1) is correct.

SECTION - E

Matrix-Match Type Questions

1. Let $f(x)$ denote the determinant

$$f(x) = \begin{vmatrix} x^2 & 2x & 1+x^2 \\ x^2+1 & x+1 & 1 \\ x & -1 & x-1 \end{vmatrix}$$

on expansion $f(x)$ is seen to be a 4th degree polynomial given by

$$f(x) = a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4.$$

Using differentiation of determinant or otherwise, match the values of the quantities on the left to those on the right.

Column-I

- (A) a_4
(B) a_3
(C) a_0
(D) $a_1^2 + a_1 + 1$

Column-II

- (p) -1
(q) 1
(r) -3
(s) 3

Sol. Answer A(p), B(q), C(r), D(s)

$$\text{If } x = 0, a_4 = \begin{vmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & -1 & -1 \end{vmatrix} = -1$$

$$\begin{aligned} f(x) &= x^2[(x^2 - 1) - 1] - 2x[(x^2 + 1)(x - 1) - x] + (1 + x^2)[(1 + x^2)(-1) - x(x + 1)] \\ &= x^2(x^2 - 2) - 2x(x^3 - x^2 - 1) - (1 + x^2)^2 - x(1 + x + x^2 + x^3) \\ &= (x^4 - 2x^2) - (2x^4 - 2x^3 - 2x) - (1 + 2x^2 + x^4) - (x + x^2 + x^3 + x^4) \\ &= -3x^4 + x^3 - 5x^2 + x - 1 \end{aligned}$$

$$\therefore a_4 = -1, a_3 = 1, a_0 = -3, a_1 = 1$$

$$\therefore a_1^2 + a_1 + 1 = 3$$

2. If A is a non-singular matrix of order $n \times n$, then match the following

Column-I

- (A) $(\text{adj } A)^{-1}$
(B) $\text{adj } (KA)$
(C) $\text{adj } (\text{adj } A)$
(D) $\text{adj } (A^{-1})$

Column-II

- (p) $k^n (\text{adj } A)$
(q) $\frac{A}{|A|}$
(r) $|A|^{n-2} A$
(s) $\frac{\text{adj } (\text{adj } A)}{|A|^{n-1}}$
(t) $k^{n-1} (\text{adj } A)$

Sol. Answer A(q, s), B(t), C(r), D(q, s)

We know that $(\text{adj } A)(A) = A(\text{adj } A) = |A|I_n$ and $\text{adj}(\text{adj } A) = |A|^{n-2}A$; $|A| \neq 0$

3. The entries in a 3×3 matrix A are either 1 or -1 , then match the following

Column-I

- (A) Total number of such matrices
(B) The maximum value of $\det(A)$
(C) The maximum value of $\text{tr}(A)$
(D) The minimum value of $\det(A)$

Column-II

- (p) 4
(q) 3
(r) 512
(s) Zero
(t) -4

Sol. Answer A(r), B(p), C(q), D(t)

(A) $2^9 = 512$

(B) For maximum value the determinant will be

$$\begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ -1 & -1 & 1 \end{vmatrix} = 1(1+1) + 1(1+1) + 0 = 4$$

(C) Maximum trace = $1 + 1 + 1 = 3$

(D) Minimum value = -4

SECTION - F

Integer Answer Type Questions

1. Let $|A| = |a_{ij}|_{3 \times 3} \neq 0$. Each element a_{ij} is multiplied by k^{-1} . Let $|B|$ be the resulting determinant, where $k_1|A| + k_2|B| = 0$, then $k_1 + k_2$ is

Sol. Answer (0)

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$|B| = \begin{vmatrix} a_{11} & k^{-1}a_{12} & k^{-2}a_{13} \\ k a_{21} & a_{22} & k^{-1}a_{23} \\ k^2 a_{31} & k a_{32} & a_{33} \end{vmatrix} = \frac{1}{k^3} \begin{vmatrix} k^2 a_{11} & k a_{12} & a_{13} \\ k^2 a_{21} & k a_{22} & a_{23} \\ k^2 a_{31} & k a_{32} & a_{33} \end{vmatrix} = |A|$$

$$\Rightarrow |A| = |B|$$

$$\Rightarrow |A| - |B| = 0$$

Comparing it with $k_1|A| + k_2|B| = 0$

We get $k_1 + k_2 = 0$

2. If $\begin{vmatrix} x^2 - 6x + 5 & 2x - 6 & 2 \\ 2x^2 - 5x + 9 & 4x - 5 & 4 \\ 6x^2 + 4x + 6 & 12x + 4 & 12 \end{vmatrix} = Ax^3 + Bx^2 + Cx + D$, then $A + B + 2C$ is equal to

Sol. Answer (0)

As the value of the determinant is zero

$$\Rightarrow A = B = C = D = 0$$

$$\Rightarrow A + B + 2C + 3D = 0$$

3. Let $A = [a_{ij}]_{n \times n}$, n is odd. Then $\det((A - A^T)^{2009})$ is equal to

Sol. Answer (0)

$\therefore A - A^T$ is skew symmetric.

$\Rightarrow (A - A^T)^{2009}$ is also skew symmetric

$\Rightarrow \det(A - A^T)^{2009} = 0$, as determinant value of every skew symmetric matrix of odd order is 0.

4. If $\Delta = \begin{vmatrix} 1 & 3\cos\theta & 1 \\ \sin\theta & 1 & 3\cos\theta \\ 1 & \sin\theta & 1 \end{vmatrix}$, then the $\frac{1}{1000}$ [maximum value of Δ – minimum value of Δ]³ is equal to _____.

Sol. Answer (1)

$$\begin{aligned} \therefore \Delta &= 1\{1 - 3\sin\theta\cos\theta\} - 3\cos\theta\{\sin\theta - 3\cos\theta\} + 1\{\sin^2\theta - 1\} \\ &= 1 - 6\sin\theta\cos\theta + 9\cos^2\theta + \sin^2\theta - 1 \end{aligned}$$

$$\therefore \Delta = (\sin\theta - 3\cos\theta)^2$$

$$\therefore -\sqrt{10} \leq \sin\theta - 3\cos\theta \leq \sqrt{10}$$

$$\therefore 0 \leq (\sin\theta - 3\cos\theta)^2 \leq 10$$

$$\Rightarrow 0 \leq \Delta \leq 10$$

$$\therefore \frac{1}{100} [\text{maximum value of } \Delta - \text{minimum value of } \Delta]^3$$

$$= \frac{1}{1000} [(10 - 0)^3] = \frac{1000}{1000} = 1$$

5. Given $2x + 4y + z = 1$, $\lambda x + 2y + z = 2$, $x + y - \lambda z = 3$, then one of the value of λ such that the given system of equations has no solution, is _____.

Sol. Answer (0)

Since, given system has no solution,

$\therefore \Delta = 0$ and any one amongst Δ_x , Δ_y , Δ_z is non-zero.

$$\text{Let } \begin{vmatrix} 2 & 4 & 1 \\ \lambda & 2 & 1 \\ 1 & 1 & -\lambda \end{vmatrix} = 0 \quad \Rightarrow 4\lambda^2 - 3\lambda = 0$$

$$\Rightarrow \lambda(4\lambda - 3) = 0$$

$$\Rightarrow \lambda = 0, \frac{3}{4}$$

$$\text{and } \Delta_x = \begin{vmatrix} 1 & 4 & 1 \\ 2 & 2 & 1 \\ 3 & 1 & -\lambda \end{vmatrix} \neq 0$$

6. Let (x, y, z) be points with integer coordinates satisfying the system of homogeneous equations:

$$3x - y - z = 0$$

$$-3x + z = 0$$

$$-3x + 2y + z = 0$$

Then the number of such points for which $x^2 + y^2 + z^2 \leq 100$ is

[IIT-JEE 2009]

Sol. Answer (7)

We have

$$3x - y - z = 0 \quad \dots(i)$$

$$-3x + z = 0 \quad \dots(ii)$$

$$-3x + 2y + z = 0 \quad \dots(iii)$$

Applying (i) and (ii) we get

$$\Rightarrow y = 0 \quad \dots(iv)$$

Also $3x = z$

Points satisfying $x^2 + y^2 + z^2 \leq 100$ with integral coordinates can be

$$(0, 0, 0), (1, 0, 3), (2, 0, 6), (3, 0, 9), (-1, 0, -3), (-2, 0, -6), (-3, 0, -9)$$

Hence 7 such points exist.

7. Let ω be the complex number $\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$. Then the number of distinct complex numbers z satisfying

$$\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0$$
 is equal to

[IIT-JEE 2010]

Sol. Answer (1)

$$\text{Let } A = \begin{bmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \text{Tr}(A) = 0, |A| = 0$$

$$\therefore A^3 = 0$$

$$\Rightarrow \begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = |A + zI| = 0$$

$$\Rightarrow z^3 = 0$$

$$\Rightarrow z = 0, \text{ the number of } z \text{ satisfying the given equation is } 1.$$

8. The number of all possible values of θ , where $0 < \theta < \pi$, for which the system of equations

$$(y + z)\cos 3\theta = (xyz)\sin 3\theta$$

$$x \sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$$

$$(xyz)\sin 3\theta = (y + 2z)\cos 3\theta + y\sin 3\theta$$

have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is

[IIT-JEE 2010]

Sol. Answer (3)

Given equation can be written as

$$x \sin 3\theta - \frac{\cos 3\theta}{y} - \frac{\cos 3\theta}{z} = 0 \quad \dots(i)$$

$$x \sin 3\theta - \frac{2\cos 3\theta}{y} - \frac{2\sin 3\theta}{z} = 0 \quad \dots(ii)$$

$$x \sin 3\theta - \frac{2}{y} \cos 3\theta - \frac{1}{z} (\cos 3\theta + \sin 3\theta) = 0 \quad \dots(iii)$$

Equations (ii) and (iii) imply

$$2\sin 3\theta = \cos 3\theta + \sin 3\theta$$

$$\Rightarrow \sin 3\theta = \cos 3\theta$$

$$\therefore \tan 3\theta = 1 \Rightarrow 3\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}$$

$$\text{or } \theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12}$$

9. Let k be a positive real number and let $A = \begin{bmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-2k & 0 & 2\sqrt{k} \\ -\sqrt{k} & -2\sqrt{k} & 0 \end{bmatrix}$.

If $\det(\text{adj } A) + \det(\text{adj } B) = 10^6$, then $[k]$ is equal to

[IIT-JEE 2010]

[Note : $\text{adj } M$ denotes the adjoint of a square matrix M and $[k]$ denotes the largest integer less than or equal to k]

Sol. Answer (4)

$$|A| = (2k + 1)^3, |B| = 0$$

$$\Rightarrow \det(\text{adj } A) \neq \det(\text{adj } B) = (2k + 1)^6 = 10^6 \Rightarrow k = \frac{9}{2}.$$

$$\Rightarrow [k] = 4$$

10. Let M be a 3×3 matrix satisfying $M \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$, $M \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$, and $M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$

Then the sum of the diagonal entries of M is

[IIT-JEE 2011]

Sol. Answer (9)

Let $M = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ be the given matrix.

Using the given conditions, we have

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$$

$$\Rightarrow a_{12} = -1$$

$$a_{22} = 2$$

$$a_{32} = 3$$

$$\text{Also, } M \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow a_{11} - a_{12} = 1$$

$$a_{21} - a_{22} = 1$$

$$a_{31} - a_{32} = -1$$

Using above equations, we shall get

$$a_{11} = 0$$

$$\text{Moreover, } M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$$

$$\Rightarrow a_{11} + a_{12} + a_{13} = 0$$

$$a_{21} + a_{22} + a_{23} = 0$$

$$a_{31} + a_{32} + a_{33} = 12$$

Using above results, we get

$$a_{33} = 7$$

Finally, the sum of elements of leading diagonals

$$= a_{11} + a_{22} + a_{33}$$

$$= 0 + 2 + 7$$

$$= 9$$

11. The total number of distinct $x \in \mathbb{R}$ for which $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ is

[JEE(Advanced)-2016]

Sol. Answer (2)

$$\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$$

$$x^3 \begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 3 & 9 & 1 \end{vmatrix} + x^6 \begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & 8 \\ 3 & 9 & 27 \end{vmatrix} = 10$$

$$R_3 \rightarrow R_3 - R_2 \quad C_3 \rightarrow C_3 - C_2$$

$$R_2 \rightarrow R_2 - R_1 \quad C_2 \rightarrow C_2 - C_1$$

$$x^3 \begin{vmatrix} 1 & 1 & 1 \\ 1 & 3 & 0 \\ 1 & 5 & 0 \end{vmatrix} + x^6 \begin{vmatrix} 1 & 0 & 0 \\ 2 & 2 & 4 \\ 3 & 6 & 18 \end{vmatrix}$$

$$2x^6 + 2x^3 - 10 = 0$$

$$6x^6 + x^3 - 5 = 0$$

$$(x^3 + 1)(6x^3 - 5) = 0$$

12. Let P be a matrix of order 3×3 such that all the entries in P are from the set $\{-1, 0, 1\}$. Then, the maximum possible value of the determinant of P is _____. [JEE(Advanced)-2018]

Sol. Answer (4.00)

$$P = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

The maximum possibility of $|P|$ can be 6 if

$$\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} = \pm 2 = \pm \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = \pm \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

but, the moment $\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$ is set as 2 (say) and $\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$ as 2 or -2 , automatically $\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$ takes zero.

Hence, $|P| \neq 6$. Next possibility is 4.

$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} \text{ is one such possibility.}$$

SECTION - G

Multiple True-False Type Question

1. The matrices which commute with $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ in case of multiplication

STATEMENT-1 : Are always singular.

STATEMENT-2 : Are always non-singular.

STATEMENT-3 : Are always symmetric.

(1) F F F

(2) T T F

(3) T T T

(4) T F T

Sol. Answer (1)

Let $B = \begin{bmatrix} x & y \\ a & b \end{bmatrix}$ commute with A

$$\Rightarrow AB = BA$$

$$\text{i.e., } \begin{bmatrix} x+a & y+b \\ a & b \end{bmatrix} = \begin{bmatrix} x & x+y \\ a & a+b \end{bmatrix}$$

$$\therefore x+a = x, y+b = x+y, b = a+b$$

$$\Rightarrow a = 0 \text{ and } b = x$$

$$\therefore B = \begin{bmatrix} x & y \\ 0 & x \end{bmatrix}; \text{ option (1) is correct.}$$

SECTION - H

Aakash Challengers Questions

1. If the system of equation $ax + by + cz = 0$, $bx + cy + az = 0$, $cx + ay + bz = 0$ has non-trivial solution then

find the value of $\begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix}$.

Sol. Answer (0)

For non-trivial solution

$$A = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$$

And the determinant

$$\begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix} \text{ is determinant of cofactor matrix of } A.$$

Hence its value = 0

2. If a, b, c are real such that $a^2 + b^2 + c^2 = 1$ then show that $\begin{vmatrix} ax-by-c & bx+ay & cx+a \\ bx+ay & -ax+by-c & cy+b \\ cx+a & cy+b & -ax-by+c \end{vmatrix} = 0$ will represent a straight line.

Sol. Applying $C_1 \rightarrow aC_1 + bC_2 + cC_3$

$$\Delta = \begin{vmatrix} (a^2+b^2+c^2)x & bx+ay & cx+a \\ (a^2+b^2+c^2)y & -ax+by-c & cy+b \\ (a^2+b^2+c^2) & cy+b & -ax-by+c \end{vmatrix}$$

$$\Delta = \begin{vmatrix} x & bx+ay & cx+a \\ y & -ax+by-c & cy+b \\ 1 & cy+b & -ax-by+c \end{vmatrix} \because a^2+b^2+c^2=1$$

$$C_2 \rightarrow C_2 - bC_1, C_3 \rightarrow C_3 - cC_1$$

$$\Delta = \begin{vmatrix} x & ay & a \\ y & -ax-c & b \\ 1 & cy & -ax-by \end{vmatrix}$$

$$R_3 \rightarrow xR_1 + yR_2 + R_3$$

$$\Delta = \begin{vmatrix} x & ay & a \\ y & -ax-c & b \\ x^2+y^2+1 & 0 & 0 \end{vmatrix}$$

Expand w.r.t. R_3

$$\Delta = (x^2 + y^2 + 1) [aby + a^2x + ac]$$

$$= a(x^2 + y^2 + 1) (ax + by + c)$$

$\Delta = 0 \Rightarrow ax + by + c = 0$ represents a straight line.

3. Find the value of $\begin{vmatrix} 1! & 2! & 3! \\ 2! & 3! & 4! \\ 3! & 4! & 5! \end{vmatrix}$.

Sol. Given $\begin{vmatrix} 1! & 2! & 3! \\ 2! & 3! & 4! \\ 3! & 4! & 5! \end{vmatrix}$

$$= (2!)(3!) \begin{vmatrix} 1 & 2 & 6 \\ 1 & 3 & 12 \\ 1 & 4 & 20 \end{vmatrix}$$

$$R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_3$$

$$= 12 \begin{vmatrix} 0 & -1 & -6 \\ 0 & -1 & -8 \\ 1 & 4 & 20 \end{vmatrix}$$

Expanding along C_1 , we get

$$= 12((-1)(-8) - (-1)(-6))$$

$$= 24$$

4. Show that

$$\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ac - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix} = \begin{vmatrix} bc - a^2 & ca - b^2 & ab - c^2 \\ ca - b^2 & ab - c^2 & bc - a^2 \\ ab - c^2 & bc - a^2 & ca - b^2 \end{vmatrix}$$

Sol. A direct evaluation of the determinants involved is not at all attractive. One is required to think out of the box to discover what may be the idea behind the solution.

Write LHS as product of two determinants

$$\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ac - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix} = \begin{vmatrix} -a^2 + bc + bc & -ab + ab + c^2 & -ac + b^2 + ca \\ -ab + c^2 + ba & -b^2 + ac + ca & -bc + bc + a^2 \\ -ac + ca + b^2 & -bc + a^2 + bc & -c^2 + ab + ab \end{vmatrix}$$

$$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} -a & c & b \\ -b & a & c \\ -c & b & a \end{vmatrix} \quad (\text{Row-by-row multiplication})$$

$$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}^2$$

$$\text{i.e., } \begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ca - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix} = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}^2 \quad \dots(A)$$

Now how to relate the determinant $\begin{vmatrix} bc - a^2 & ca - b^2 & ab - c^2 \\ ca - b^2 & ab - c^2 & bc - a^2 \\ ab - c^2 & bc - a^2 & ca - b^2 \end{vmatrix}$ to the determinant $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}^2$?

$$\text{Consider } \Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

The determinant formed by cofactors of elements of Δ is

$$\Delta_1 = \begin{vmatrix} bc - a^2 & ca - b^2 & ab - c^2 \\ ca - b^2 & ab - c^2 & bc - a^2 \\ ab - c^2 & bc - a^2 & ca - b^2 \end{vmatrix}$$

and we know that $\Delta_1 = \Delta^2$

$$\text{Thus } \begin{vmatrix} bc - a^2 & ca - b^2 & ab - c^2 \\ ca - b^2 & ab - c^2 & bc - a^2 \\ ab - c^2 & bc - a^2 & ca - b^2 \end{vmatrix} = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}^2 \quad \dots(B)$$

From (A) and (B) we have

$$\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ca - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix} = \begin{vmatrix} bc - a^2 & ca - b^2 & ab - c^2 \\ ca - b^2 & ab - c^2 & bc - a^2 \\ ab - c^2 & bc - a^2 & ca - b^2 \end{vmatrix}$$

$$5. \quad \text{Let } \begin{vmatrix} x^2 + 3x & x - 1 & x + 3 \\ x + 1 & -2x & x - 4 \\ x - 3 & x + 4 & 3x \end{vmatrix} = ax^4 + bx^3 + cx^2 + dx + k$$

where a, b, c, d, k are independent of x . Find the value of $4a + 3b + 2c + d$.

Sol. Rather than expand the determinant, we differentiate both sides w.r.t. x and then put $x = 0$ to obtain the value of d .

$$\Delta(x) = \begin{vmatrix} x^2 + 3x & x - 1 & x + 3 \\ x + 1 & -2x & x - 4 \\ x + 3 & x + 4 & 3x \end{vmatrix} = ax^4 + bx^3 + cx^2 + dx + k$$

Differentiating w.r.t. x

$$\Delta'(x) = \begin{vmatrix} 2x+3 & 1 & 1 \\ x+1 & -2 & 1 \\ x-3 & x+4 & 3x \end{vmatrix} + \begin{vmatrix} x^2+3x & x-1 & x+3 \\ 1 & -2 & 1 \\ x-3 & x+4 & 3x \end{vmatrix} + \begin{vmatrix} x^2+3x & x-1 & x+3 \\ x+1 & -2x & x-4 \\ 1 & 1 & 3 \end{vmatrix}$$

$$= 4ax^3 + 3bx^2 + 2cx + d$$

Set $x = 1$ on both sides to obtain

Then, the value of $4a + 3b + 2c + d = 15$.

Remarks :

(1) If the value of k (the constant term) is to be found out, just put $x = 0$ in the determinant.

(2) If the value of b is to be found out, then divide both sides by x^4 and write the polynomial in t , where $t = \frac{1}{x}$. Now differentiate, as in the example above, w.r.t. t to find the desired coefficient.

6. In the determinant $\begin{vmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ 2 & -3 & 0 \end{vmatrix}$, the ratio of co-factor and minor of element -3 is
- (1) -1 (2) 0 (3) 1 (4) 2

Sol. Answer (1)

Ratio of cofactor to its minor of element -3 , which is in 3rd row and 2nd column $= (-1)^{3+2} = -1$

7. $\begin{vmatrix} 1 & 5 & \pi \\ \log_e e & 5 & \sqrt{5} \\ \log_{10} 10 & 5 & e \end{vmatrix}$ is equal to

- (1) $\sqrt{\pi}$ (2) e (3) 1 (4) 0

Sol. Answer (4)

$$\begin{vmatrix} 1 & 5 & \pi \\ 1 & 5 & \sqrt{5} \\ 1 & 5 & e \end{vmatrix} = 5 \begin{vmatrix} 1 & 1 & \pi \\ 1 & 1 & \sqrt{5} \\ 1 & 1 & e \end{vmatrix} = 0$$

8. If a, b, c are in A.P., then $\begin{vmatrix} x+2 & x+3 & x+a \\ x+4 & x+5 & x+b \\ x+6 & x+7 & x+c \end{vmatrix}$ is

- (1) $x - (a + b + c)$ (2) $9x^2 + a + b + c$ (3) $a + b + c$ (4) 0

Sol. Answer (4)Apply $C_2 \rightarrow C_2 - C_1$

$$\therefore \begin{vmatrix} x+2 & 1 & x+a \\ x+4 & 1 & x+b \\ x+6 & 1 & x+c \end{vmatrix} \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - R_1} \begin{vmatrix} x+2 & 1 & x+a \\ 2 & 0 & b-a \\ 4 & 0 & c-a \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} x+2 & 1 & x+a \\ 2 & 0 & b-a \\ 4 & 0 & c-a \end{vmatrix} = 1[2(c-a) - 4(b-a)] = 2\{2b - c - a\}$$

$$\therefore \Delta = 0 \quad \{\because 2b = a + c\}$$

9. If $\begin{vmatrix} 1+ax & 1+bx & 1+cx \\ 1+a_1x & 1+b_1x & 1+c_1x \\ 1+a_2x & 1+b_2x & 1+c_2x \end{vmatrix} = A_0 + A_1x + A_2x^2 + A_3x^3$, then A_0 is

(1) abc

(2) 0

(3) 1

(4) -1

Sol. Answer (2)

Put $x = 0$ on both sides $\Rightarrow A_0 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0$

10. Number of real roots of equation $\begin{vmatrix} x^2 & -4 & -2 \\ 4 & x^2 & 1 \\ -5 & 3 & x^2 \end{vmatrix} = 0$ are

(1) 0

(2) 1

(3) 2

(4) 4

Sol. Answer (3)

Given $\begin{vmatrix} x^2 & -4 & -2 \\ 4 & x^2 & 1 \\ -5 & 3 & x^2 \end{vmatrix} = 0$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$\Rightarrow \begin{vmatrix} x^2 - 1 & x^2 - 1 & x^2 - 1 \\ 4 & x^2 & 1 \\ -5 & 3 & x^2 \end{vmatrix} = 0$$

$$\Rightarrow (x^2 - 1) \begin{vmatrix} 1 & 1 & 1 \\ 4 & x^2 & 1 \\ -5 & 3 & x^2 \end{vmatrix} = 0$$

Expanding along R_1 ,

$$\Rightarrow (x^2 - 1)[x^4 - 3 - 5 - 4x^2 + 12 + 5x^2] = 0$$

$$\Rightarrow (x^2 - 1)(x^4 + x^2 + 4) = 0$$

$$\Rightarrow x^2 = 1, \text{ i.e., } x = \pm 1$$

11. The coefficient of x in $f(x) = \begin{vmatrix} x & 1+\sin x & \cos x \\ 1 & \log(1+x) & 2 \\ x^2 & 1+x^2 & 0 \end{vmatrix}$ where $-1 < x \leq 1$, is

- (1) 1 (2) -2 (3) -1 (4) 0

Sol. Answer (2)

$$f(x) = x\{-2(1+x^2)\} - (1+\sin x)(-2x^2) + \cos x\{1+x^2 - x^2\log(1+x)\}$$

$$= -2x - 2x^3 + 2x^2 + 2x^2\sin x + \cos x\{1+x^2 - x^2\log(1+x)\}$$

Hence, -2

12. If $\text{adj}B = A$, $|P| = |Q| = 1$, then $\text{adj}(Q^{-1}BP^{-1})$ equals

- (1) PQ (2) QAP (3) PAQ (4) $PA^{-1}Q$

Sol. Answer (3)

$$\text{adj}(Q^{-1}BP^{-1}) = \text{adj}Q^{-1} \cdot \text{adj}B \cdot \text{adj}P^{-1} = \frac{P}{|P|} A \cdot \frac{Q}{|Q|} = PAQ$$

13. The matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$ is

- (1) Orthogonal (2) Nilpotent (3) Idempotent (4) Involutory

Sol. Answer (3)

Given $A = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix} \cdot \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$$

Now as $A^2 = A$.

$\therefore A$ is idempotent matrix.

14. The matrix $A = \begin{bmatrix} a & 2 \\ 2 & 4 \end{bmatrix}$ is singular if

- (1) $a \neq 1$ (2) $a = 1$ (3) $a = 0$ (4) $a = -1$

Sol. Answer (2)

A is singular $\Rightarrow |A| = 0$

$$\therefore 4a - 4 = 0 \Rightarrow a = 1$$

15. $A = \begin{bmatrix} 1 & 0 & k \\ 2 & 1 & 3 \\ k & 0 & 1 \end{bmatrix}$ is invertible for

- (1) $k = 1$ (2) $k = -1$ (3) $k \neq \pm 1$ (4) $k = 0$

Sol. Answer (3)

A is invertible $\Rightarrow |A| \neq 0$

$$\Rightarrow K \neq \pm 1$$

16. Let a, b, c be such that $(b+c) \neq 0$ if $\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^nc \end{vmatrix} = 0$, then value of n is

- (1) 0 (2) Any even integer (3) Any odd integer (4) Any integer

Sol. Answer (3)

$$\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ a & -b & c \end{vmatrix} = \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a+1 & a-1 & a \\ b+1 & b-1 & -b \\ c-1 & c+1 & c \end{vmatrix}$$

$$= \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^{n+1} \begin{vmatrix} a+1 & a & a-1 \\ b+1 & -b & b-1 \\ c-1 & c & c+1 \end{vmatrix} \quad \{C_2 \leftrightarrow C_3\}$$

$$= [1 + (-1)^{n+2}] \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} \Rightarrow \text{any odd integer}$$

17. Prove that $\begin{vmatrix} \cos(A-P) & \cos(A-Q) & \cos(A-R) \\ \cos(B-P) & \cos(B-Q) & \cos(B-R) \\ \cos(C-P) & \cos(C-Q) & \cos(C-R) \end{vmatrix} = 0$

Sol. Given $\begin{vmatrix} \cos(A-P) & \cos(A-Q) & \cos(A-R) \\ \cos(B-P) & \cos(B-Q) & \cos(B-R) \\ \cos(C-P) & \cos(C-Q) & \cos(C-R) \end{vmatrix}$

$$\Rightarrow \begin{vmatrix} \cos A \cos P + \sin A \sin P & \cos A \cos Q + \sin A \sin Q & \cos A \cos R + \sin A \sin R \\ \cos B \cos P + \sin B \sin P & \cos B \cos Q + \sin B \sin Q & \cos B \cos R + \sin B \sin R \\ \cos C \cos P + \sin C \sin P & \cos C \cos Q + \sin C \sin Q & \cos C \cos R + \sin C \sin R \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} \cos A & \sin A & 0 \\ \cos B & \sin B & 0 \\ \cos C & \sin C & 0 \end{vmatrix} \cdot \begin{vmatrix} \cos P & \sin P & 0 \\ \cos Q & \sin Q & 0 \\ \cos R & \sin R & 0 \end{vmatrix} = 0$$

